AEC-YOLO: An Adaptive Edge-Aware Calibration Framework for Robust Small Object Detection in Low-Altitude Remote Sensing Imagery

Yadong Liu, Shengbang Zhou, Chuanqi Li, Dong Chen, Shutian Liu, Yuhua Zhang, and Yixing Gao

***Abstract*—Small object detection in uncrewed aerial vehicle (UAV) low-altitude remote sensing imagery faces core challenges, specifically edge information degradation and feature misalignment resulting from complex backgrounds and drastic scale variations. To address these issues, this study introduces AEC-YOLO. This framework aims to actively preserve and refine edge information. Based on the YOLO11s architecture, the proposed model achieves performance improvements through synergistic module optimization. First, a parallel edge refinement architecture (PERA) is integrated into the backbone network to actively extract and reinforce high-frequency edge details. Second, an adaptive edge calibration mechanism (AECM) is designed in the neck network. Specifically, a bidirectional attention calibration network (BAC-Net) facilitates the collaborative optimization of deep semantics and shallow edge features. In addition, a P2 detection head is introduced to enhance the response to small objects. Finally, an adaptive weight Focaler-MPDIoU loss function is proposed to balance the sample distribution and localization sensitivity. Experimental results on the VisDrone2019 dataset demonstrate that, compared with the baseline YOLO11s, AEC-YOLO improves mAP_{50} by 9.43 percentage points and $mAP_{50:95}$ by 6.45 percentage points. Furthermore, to verify the generalization capability, the framework was extensively evaluated on the AI-TOD, DOTA-v1.5, and DIOR datasets, achieving mAP_{50} improvements of 4.9, 4.6, and 4.96 percentage points over the baseline, respectively. Comparative evaluations with state-of-the-art methods further demonstrate the superior performance and robustness of the proposed framework in complex low-altitude scenarios. The results indicate that the preservation of edge information and the multiscale feature collaborative calibration mechanism significantly elevate small object detection accuracy, providing a new technical path for UAV visual perception.**

***Index Terms*—Adaptive loss function, aerial imagery, edge refinement, feature calibration, small object detection.**

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I. Introduction

As a low-altitude remote sensing platform, the uncrewed aerial vehicle (UAV) demonstrates significant application value in the field of Earth observation, owing to its high mobility, wide-area observation capabilities, and low deployment costs. This technology has been widely applied in critical scenarios, such as smart city management [1], precision agriculture [2], public safety [3], and infrastructure inspection [4], significantly reshaping the operational paradigms of related industries. However, the unique aerial observation perspective also imposes severe physical and environmental challenges on small object detection. First, limited by the physical constraints of ground sampling distance (GSD), ground objects exhibit pixel-level representations under aerial imaging conditions. Moreover, the dense distribution of targets is prone to cause spatial occlusion effects. Second, the image quality is frequently degraded by atmospheric scattering. When compounded by motion blur from high-speed platform movement, key spectral, and spatial features suffer severe degradation. In addition, the low contrast problem caused by clutter interference and shadow effects in complex ground object scenes, as well as the sharp fluctuations in target scale due to dynamic changes in flight altitude and angle, all constitute the main technical bottlenecks for improving the performance of current object detection algorithms.

The development of object detection technology has evolved from early machine learning methods based on handcrafted features (such as histogram of oriented gradients (HOG) [5]) and sliding windows to deep learning-driven paradigms. While the breakthrough of the Viola-Jones algorithm [6] in 2001 improved efficiency in face detection, its application scope remained limited. In 2012, the success of AlexNet in the ImageNet competition facilitated the widespread adoption of deep learning in computer vision [7]. Subsequently, in 2015, Faster R-CNN [8] further implemented end-to-end training through the region proposal network (RPN), significantly improving speed and accuracy. The emergence of the you only look once (YOLO) [9] series in 2016 established the one-stage detection framework, realizing efficient real-time detection through a single forward propagation. Recently, models such as the detection Transformer (DETR) [10] have utilized self-attention mechanisms to handle global context, offering new approaches for addressing scale variation and occlusion.

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To address the challenges of small object detection from UAV perspectives, advanced single-stage detectors, represented by the YOLO [9] series, have been widely adopted due to their superior balance between speed and accuracy. However, these models designed for general scenarios have exposed inherent structural bottlenecks when dealing with drone aerial images. The downsampling mechanism of their standard convolutional networks inevitably leads to the loss of high-frequency details such as edges [11], [12], [13]. Moreover, in the feature pyramid structure, this loss will exacerbate the semantic gap between deep semantic features and shallow edge features, resulting in insufficient feature alignment. Consequently, deep semantics neither effectively guide the precise localization of shallow details nor can shallow edge features sufficiently compensate for the ambiguity of deep semantics. This misalignment severely restricts the boundary positioning accuracy of small objects. Given that distinguishing small objects from complex backgrounds relies heavily on edge contours as critical discriminative cues, the passive degradation of edges during feature extraction and the subsequent feature misalignment constitute the core bottlenecks restricting detection performance. Therefore, this study focuses on the active protection and enhancement mechanisms of edge information.

Based on these insights, this article proposes AEC-YOLO, utilizing YOLO11s as the baseline. Specifically, the parallel edge refinement architecture (PERA) structure is integrated into the backbone network to reinforce high-frequency details, such as edges, at the source. In the neck network, the AECM is designed to intelligently fuse deep semantics and shallow edge details via bidirectional attention calibration network (BAC-Net). Furthermore, a dedicated P2 detection head is introduced to further enhance small object detection performance. Simultaneously, an adaptive weight Focaler-MPDIoU loss function is formulated to mitigate sample imbalance and localization sensitivity issues, thereby significantly improving the precision and robustness of small object detection.

The main contributions of this article are summarized as follows.

1) The PERA structure is proposed. Through parallel branches, edge contours are extracted and sharpened at low-resolution semantic levels. By combining a progressive pyramid strategy with the dual selection mechanism (DSM) [14], hierarchical fusion and task-oriented screening of multiscale edge information are achieved. This effectively addresses the loss of high-frequency edge details caused by downsampling and generates feature representations where semantic features and boundary details are highly coupled.
2) The AECM is designed. Its core component, BAC-Net, utilizes bidirectional attention and gating mechanisms to replace traditional feature pyramid concatenation operations. This facilitates the guidance of shallow edges by deep semantics and the sharpening of deep semantics by shallow edges, markedly reducing the semantic gap problem. Furthermore, a dedicated P2 detection head is introduced to reinforce small object detection performance.
3) The adaptive weight Focaler-MPDIoU loss function is proposed. It integrates the Focaler-IoU mechanism [15] with the minimum point distance IoU loss (MPDIoU Loss) [16]. By dynamically scheduling fusion weights based on the average intersection over union ($\overline{\text{IoU}}$) of the training batch, sample imbalance and localization sensitivity are adaptively optimized, thereby accelerating the refinement and convergence efficiency of model training.
4) The AEC-YOLO framework is constructed by systematically integrating the aforementioned innovations. Validations on the VisDrone2019 [17], AI-TOD [18], DOTA-v1.5 [19], and DIOR [20] datasets demonstrate that its detection accuracy outperforms the baseline model and numerous existing methods, achieving superior performance.

II. Related Works

A. General Object Detection

Object detection aims to identify the locations and categories of objects in images. Breakthroughs in deep learning have driven the transition from two-stage to one-stage detectors, focusing on the balance between accuracy and inference efficiency. Two-stage methods are centered on region-based convolutional neural networks (R-CNNs) [21] for convolutional neural network (CNN) feature extraction, Fast R-CNN [22] for shared region of interest (RoI) pooling, and Faster R-CNN [8] for integrated RPN end-to-end training, while incorporating residual network (ResNet) and feature pyramid network (FPN) to enhance multiscale accuracy. One-stage detectors prioritize real-time performance, exemplified by the YOLO series. YOLOv1 [9] pioneered the single forward pass, while YOLOv2 [23]/v3 [24] and Single Shot MultiBox Detector (SSD) optimized anchor mechanisms to improve small object detection. Subsequent versions (YOLOv5 to YOLOv13 [25], [26], [27], [28], [29], [30], [31], [32], [33]) achieve efficient detection through multitask learning, nonmaximum suppression (NMS)-free designs, and architectural innovations.

Regarding loss functions, design has evolved from early simple regression to deep optimization focusing on intersample relationships and localization accuracy. RetinaNet [34] introduced focal loss to address sample imbalance. Ranking losses, such as rank and sort loss (RS Loss) [35], optimize the rank of predicted boxes to improve quality, while distribution-aware losses, such as generalized focal loss (GFL Loss) [36], model bounding box distributions to capture the localization uncertainty. Although these methods perform excellently on general datasets, small objects in UAV scenarios occupy minimal pixel areas, lack clear textures, and are sparsely distributed. Under extreme conditions, complex distribution modeling tends to introduce noise, and ranking mechanisms struggle to maintain robust gradients amidst background interference.

Simultaneously, anchor-free frameworks, such as CenterNet [37] and DETR [10], utilize keypoint regression and Transformers to reshape application paradigms. However, existing methods still face core bottlenecks of edge information degradation and feature misalignment in UAV low-altitude

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remote sensing imagery. The proposed AEC-YOLO framework significantly enhances small object detection accuracy and robustness through active edge protection, a multiscale feature calibration mechanism, and an adaptive combined loss that balances simplicity and localization accuracy.

B. Edge-Aware Feature Extraction

Edge information serves as a critical discriminative feature for distinguishing small objects from complex backgrounds, playing an irreplaceable role in small object detection within low-altitude remote sensing imagery. Recently, researchers have increasingly recognized the significance of edge information and proposed various edge-aware feature extraction methods. Qin et al. [38] introduced boundary-aware salient network (BASNet), a pioneering work that first proposed a boundary-aware salient object detection framework. By fusing boundary information with semantic features, it effectively addresses blurred boundaries and structural incompleteness. Zheng et al. [39] further developed boundary-aware network (BANet) featuring a two-stage partial decoder mechanism. It utilizes a boundary-aware decoder (BAD) and a structure-aware decoder (SAD) to guide clear edge learning, resolving nonsmooth boundaries and structural deficiencies in remote sensing detection. Yang et al. [40] proposed edge-aware network (EANet), achieving high-precision building edge detection through the collaborative design of explicit spatial resolution embedding (ESRE) and an edge extraction branch (EEB). Zeng et al. [41] proposed AESINet, which integrates a local detail aggregation module (LDAM) and a unique differential perception mechanism to effectively extract edge information amidst complex backgrounds in optical remote sensing images. Song et al. [42] developed boundary-aware feature fusion network (BAFNet), introducing a dual-stream attention mechanism and boundary-aware fusion strategies to significantly enhance the precise localization of small objects.

Motivated by these efforts, our work emphasizes the collaborative optimization of multiscale edge extraction and bidirectional calibration between deep and shallow layers. This approach improves the discriminative capability between targets and complex backgrounds, offering a systematic solution for small object detection in low-altitude remote sensing scenarios.

C. Feature Fusion and Calibration

FPN fuses deep semantics and shallow features via top-down paths to enhance multiscale representation. However, traditional fusion strategies (e.g., concatenation or elementwise addition) tend to trigger feature misalignment and information decoupling, hindering deep semantics from providing precise guidance for small object localization [43]. Consequently, researchers have shifted toward more efficient feature alignment and calibration mechanisms. Huang et al. [44] proposed feature-aligned pyramid network (FaPN), which learns pixel-level transformation offsets through a feature alignment module to resolve context inconsistency between high-level and low-level features during upsampling. Xie et al. [45] introduced feature enhancement and alignment detector (FEADet), utilizing an adaptive alignment convolution (AAC) module and an attention feature fusion (AAF) module to address the feature misalignment and aliasing effects in oriented object detection within remote sensing imagery. Deng et al. [46] proposed aggregated recalibration network (ARNet), designing a feature recalibration (FR) module to align offsets generated during feature interactions, thereby addressing detection difficulties caused by low resolution and small scales in UAV imagery.

In the frontier field of multisource spatial information calibration, LPRnet proposed by Wang et al. [47] precisely addresses the self-supervised registration challenge between LiDAR and photogrammetric point clouds via a multiscale mask learning strategy, setting a benchmark for spatial information calibration in complex scenarios. Subsequently, PiV-AHPC proposed by Jiang et al. [48] for aerial perception scenarios utilizes a pillar-voxel dual-branch encoder to collaboratively process geometric and spectral information, successfully overcoming feature distortion during the fusion process and demonstrating the significant potential of multi-dimensional collaborative perception calibration in complex tasks.

Inspired by these frontier calibration strategies and multisource feature collaboration, this study proposes the AEC-YOLO framework to address high-frequency edge degradation in 2-D remote sensing small object detection. Unlike methods focusing on spatial registration [47] or 3-D geometric-spectral calibration [48], the proposed BAC-Net employs a bidirectional attention mechanism to achieve precise calibration of boundary details by deep semantics at the 2-D pixel level.

III. Method

To address core challenges such as the degradation of object edge information and feature misalignment in UAV aerial imagery scenarios, this article presents the AEC-YOLO detection framework. This section elaborates on its architecture and key innovations across four aspects: the overall network architecture, PERA, AECM, and the adaptive weight Focaler-MPDIoU loss function.

A. Overview of the Proposed Network

Based on the YOLO11s [31] architecture, AEC-YOLO constructs an edge information enhancement system for small object detection by reconstructing the backbone network, feature pyramid, and loss function. The overall architecture is illustrated in Fig. 1. Overall, AEC-YOLO follows a logical closed loop of "edge extraction—semantic calibration—multiscale fusion—loss constraint." First, the proposed PERA is integrated into the backbone network, replacing the original C3K2 module with the C3K2_PERA structure. This achieves multiscale edge extraction across four downsampling stages, where contour features are reinforced via heterogeneous convolution kernel combinations to preserve key boundary information. Second, the FPN integrates the BAC-Net module, built upon an inverted bottleneck (IBN) structure [49], to construct the AECM. Departing from traditional upsampling and concatenation paradigms, our

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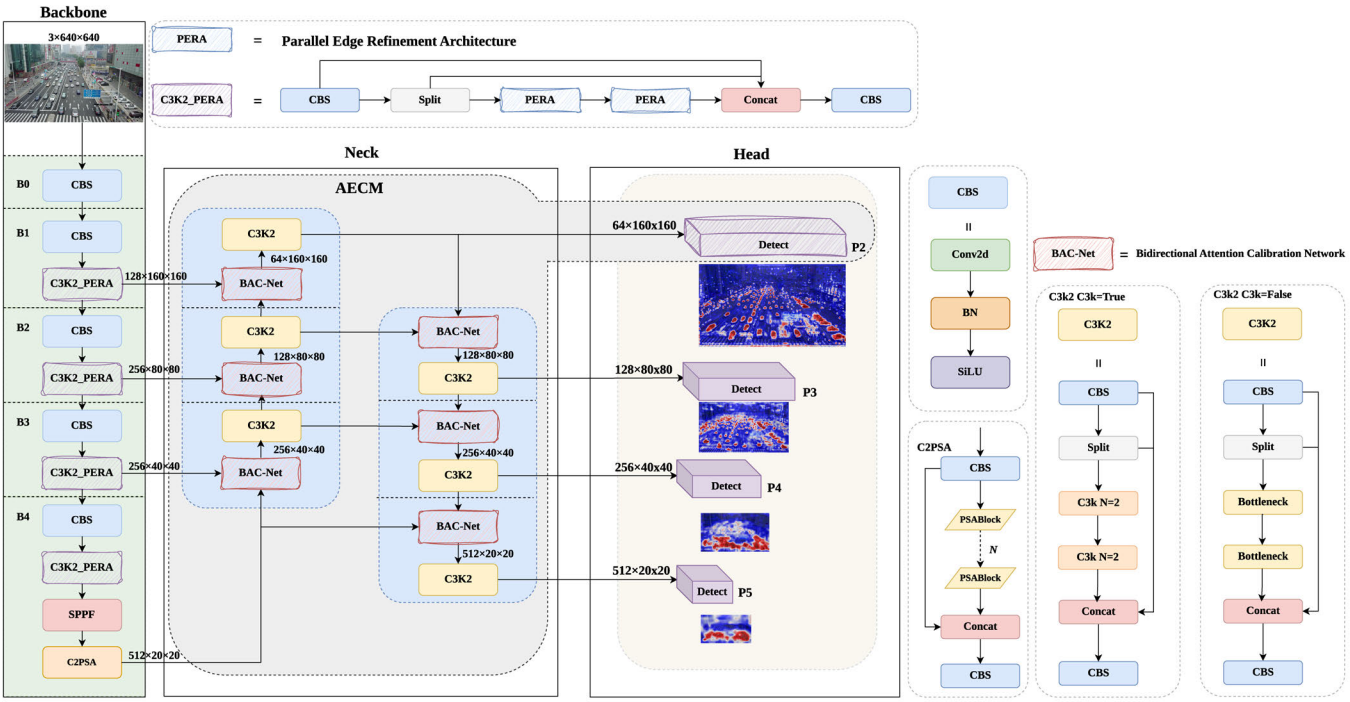


Fig. 1. Overall architecture of the proposed AEC-YOLO. The numbers annotated on the connection lines denote the dimensions of the feature maps in the format of $C \times H \times W$ (channels $\times$ height $\times$ width).

TABLE I

DETAILED CONFIGURATIONS OF THE OPERATORS AND SYMBOLS DEFINED IN THE PROPOSED ARCHITECTURE

Operator Symbol	Operation Type	Kernel Size / Stride / Padding	Channel Mapping	Normalization & Activation
$Conv_{1\times1}$	Pointwise Conv	$1\times1/1/0$	$C \to C$	BN & SiLU
$DWConv_{3\times3}$	Depthwise Conv	$3\times3/1/1$	$C \to C$	BN & SiLU
AAP_{b_i}	Adaptive Avg Pool	N/A	$C \to C$	None & None
$C_{merge,i-1}$	Pointwise Conv	$1\times1/1/0$	$2C \to C$	BN & SiLU
C_{local}	Standard Conv	$3\times3/1/1$	$C \to C$	BN & SiLU
$C_{spatial}$	Standard Conv	$3\times3/1/1$	$2 \to 1$	None & Sigmoid
C_{dw1}	Depthwise Conv	$3\times3/1/1$	$2C \to 2C$	BN & SiLU
C_{dw2}	Depthwise Conv	$3\times3/1/1$	$2C \to 2C$	BN & SiLU
C_{final}	Pointwise Conv	$1\times1/1/0$	$2C \to C$	BN & SiLU
C_{proj}	Pointwise Conv	$1\times1/1/0$	$C_{in} \to C'$	None & None
$RefineConv_L$	Depthwise Conv	$3\times3/1/1$	$C' \to C'$	BN & SiLU
$RefineConv_H$	Depthwise Conv	$3\times3/1/1$	$C' \to C'$	BN & SiLU

Note: C denotes the base channel number of the backbone network; C_{in} represents the initial channel dimension before feature fusion; C' represents the shared feature dimension after the projection layer.

framework utilizes attention gating to facilitate the reciprocal fusion of low-level edge features and high-level semantics. In addition, a P2 detection head is added to enhance responsiveness to tiny objects. Finally, the model integrates the adaptive weight Focaler-MPDIoU loss function, which dynamically adjusts fusion weights based on the average IoU metric during training. Crucially, this closed-loop architecture differs from the module stacking paradigm commonly seen in general object detection. The proposed architecture features a strict internal coupling: the high-frequency edge priors actively extracted by PERA serve as the core feature foundation for subsequent detection. Subsequently, rather than adopting a direct high-resolution scale expansion strategy, the AECM utilizes its built-in BAC-Net to guide the calibration of shallow features using deep semantics. This allows the P2 detection head to operate within a purified feature space, effectively suppressing the amplification of background noise. Finally, the dynamic loss function imposes state-aware optimization constraints tailored to this calibrated feature space. This inherent structural closed loop ensures that each component functions synergistically. Rather than being isolated insertions, they form a tightly coupled system, thereby improving the detection performance in complex aerial backgrounds. Detailed configurations of core parameterized operators (including operation types, kernel sizes, stride/padding, channel dimensions, and normalization/activation functions) are summarized in Table I.

B. PERA

Small objects within UAV perspectives occupy a minimal number of pixels in the image. Their high-frequency edge

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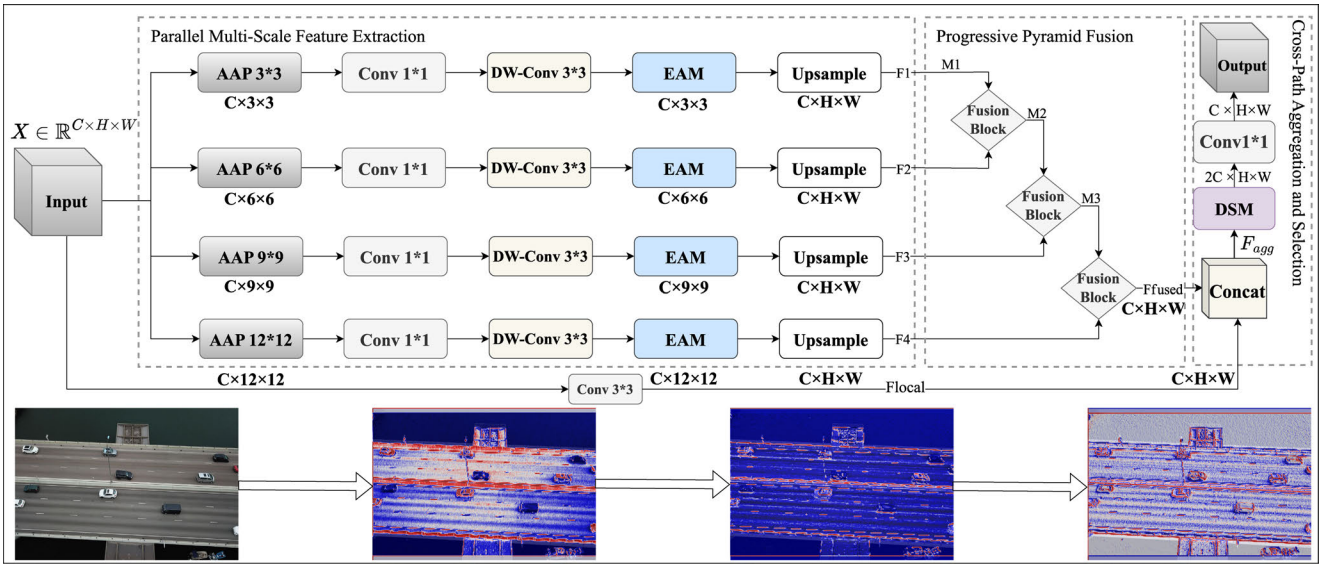


Fig. 2. Structure of PERA and the corresponding edge feature maps.

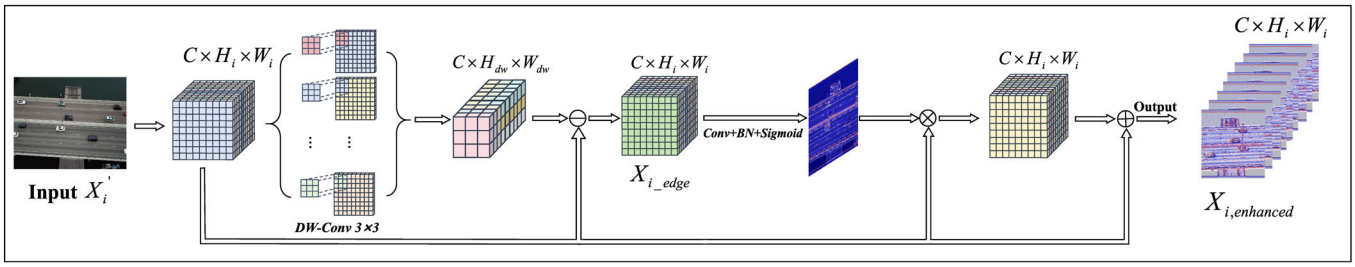


Fig. 3. Structure of EAM.

information serves as a critical cue for distinguishing them from complex backgrounds. However, standard CNNs exhibit low-pass filtering characteristics due to continuous convolution [as shown in (1)] and pooling. Consequently, this causes resolution attenuation and suppresses high-frequency signals, effectively weakening the edge contour representations of small objects [11], [12], [13]

$$\mathrm{Y}(i,j) = \sum_m \sum_n X(i-m, j-n) \cdot K(m,n) \tag{1}$$

where X and Y represent the input and output feature maps, respectively, K denotes the convolutional kernel, and (i, j) and (m, n) indicate the spatial coordinates of the feature maps and the kernel. To overcome this limitation, PERA establishes a parallel "edge information stream." It actively extracts, enhances, fuses, and screens edge features at multiple scales. Subsequently, the refined edge features are seamlessly integrated into the backbone features. Importantly, unlike conventional parallel convolutions aimed at multiscale feature enhancement, PERA is specifically designed to address the irreversible loss of high-frequency contour signals inherent in UAV imagery during downsampling. PERA actively enhances the edge gradients of objects, ensuring they are preserved before being diluted in subsequent operations. This provides an indispensable foundation for subsequent feature calibration. The specific structure of PERA and the corresponding edge feature maps are illustrated in Fig. 2.

The input to the entire process is a third-order tensor $X \in \mathbb{R}^{C\times H\times W}$. First, PERA synchronously captures multi-scale context through four parallel branches to address the drastic scale variations inherent in the UAV perspective. In each branch, the input tensor X initially undergoes spatial dimension compression via adaptive average pooling (AAP) layers of varying sizes, followed by feature transformation through pointwise convolution and depthwise convolution, respectively. Subsequently, as illustrated in Fig. 3, the core edge awareness module (EAM) applies a 3×3 depthwise convolution to the initial feature X_i' and subtracts the result from X_i' to isolate the high-frequency edge component $X_{i,edge}$. This component is processed by convolution (Conv), batch normalization (BN), and a sigmoid activation function to generate a weight map. This map acts upon the initial feature X_i' via a multiplicative gating mechanism, and the final output $X_{i,enhanced}$ is obtained through a residual connection, achieving explicit enhancement of edge information. Postprocessing, the feature maps from each branch are upsampled ($\mathcal{U}$) to the original spatial dimensions, forming a set of edge-enhanced multiscale features $\{F_1, F_2, F_3, F_4\}$. This process is formulated as follows:

$$X_i' = \mathrm{DWConv}_{3\times 3}\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{AAP}_{b_i}(X)\right)\right) \quad b_i \in \{3, 6, 9, 12\} \tag{2}$$

$$X_{i,\mathrm{edge}} = X_i' - \mathrm{DWConv}_{3\times 3}\left(X_i'\right) \tag{3}$$

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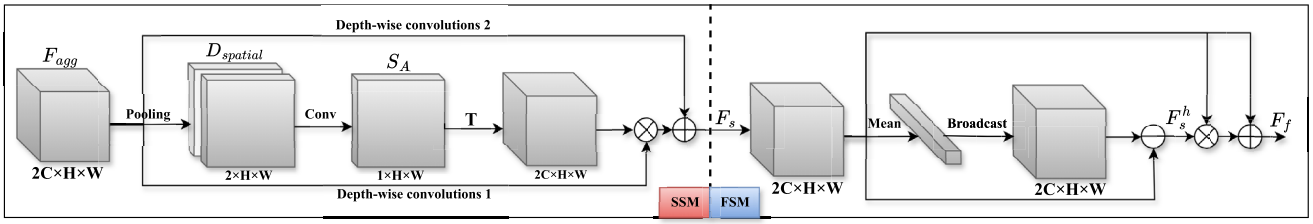


Fig. 4. Structure of the DSM. T denotes the tile function, which replicates the input $2C$ times along the channel dimension to generate $\mathbb{R}^{2C\times H\times W}$.

$$X_{i,\text{enhanced}} = X_i' + X_i' \odot \sigma\left(\text{BN}\left(\text{Conv}_{1\times 1}\left(X_{i,\text{edge}}\right)\right)\right) \quad (4)$$

$$F_i = \mathcal{U}\left(X_{i,\text{enhanced}}\right) \quad (5)$$

where b_i denotes the target output size of the pooling layer. To effectively integrate this information, PERA adopts a coarse-to-fine progressive pyramid fusion strategy. This strategy starts from the smallest scale feature F_1 and progressively concatenates and fuses larger scale features, ultimately constructing a content-rich composite edge feature F_{fused}. Each fusion module consists of a channel concatenation operation and a 1×1 convolution C_{merge} for feature integration, as shown in (6). Here, M_i denotes the intermediate feature map after fusion at the ith scale branch, M_{i-1} represents the intermediate feature map fused from the preceding $i - 1$ scale branches, with the initial state set as $M_1 = F_1$, the final fused feature F_{fused} is obtained as M_4

$$M_i = C_{merge,i-1}\left(Concat\left(M_{i-1}, F_i\right)\right) \quad i \in \{2,3,4\}. \quad (6)$$

Simultaneously, to prevent the loss of local structures during downsampling, a parallel shortcut path processes the input feature X via a 3×3 convolutional layer. This preserves the original local details to generate the local feature F_{local}. Subsequently, this feature is concatenated with the multiscale fused feature F_{fused} along the channel dimension, forming an aggregated intermediate feature $F_{agg} \in \mathbb{R}^{2C\times H\times W}$ that combines global context with local details

$$F_{local} = C_{local}\left(X\right). \quad (7)$$

To adaptively select key information from these features, the DSM [14] is introduced, as illustrated in Fig. 4. Structurally, the DSM incorporates two key submodules arranged in a series: the spatial selection module (SSM) and the subsequent frequency selection module (FSM). First, the SSM generates a two-channel spatial descriptor $D_{spatial} \in \mathbb{R}^{2\times H\times W}$ via channel pooling. Next, a convolutional layer compresses this descriptor into a single-channel spatial attention map S_A. This map dynamically modulates parallel depthwise convolution branches to fuse context information based on spatial importance, yielding the feature F_s. Subsequently, F_s is fed into the FSM for FR in the channel domain. The FSM derives the centered high-frequency feature F_s^h by removing the spatial mean within channels and uses it to enhance the original feature. This effectively amplifies the response of regions containing high-frequency details, such as object contours, resulting in F_f. The calculation process of the DSM is shown as follows:

$$S_A = C_{\text{spatial}}\left(\text{Concat}\left(\text{MaxPool}_c\left(F_{\text{agg}}\right), \text{AvgPool}_c\left(F_{\text{agg}}\right)\right)\right) \quad (8)$$

$$F_s = \left(C_{\text{dw1}}\left(F_{\text{agg}}\right)\right) \odot T\left(S_A\right) + C_{\text{dw2}}\left(F_{\text{agg}}\right) \quad (9)$$

$$F_s^h = F_s - \text{Mean}\left(F_s\right) \quad (10)$$

$$F_f = \left(F_s^h \odot F_s\right) + F_s \quad (11)$$

where the subscript c indicates pooling operations applied along the channel dimension and $T(\cdot)$ represents the tile operation to match channel dimensions. Finally, the feature maps filtered by the DSM are processed via a 1×1 convolutional layer for final feature projection and integration, yielding the final output of PERA X_{out}, as shown in (12). This output feature is infused with high-quality edge structural information derived from multiscale extraction and dual-domain screening. It establishes a superior feature foundation for subsequent network layers to achieve high-precision small object localization

$$X_{out} = C_{final}\left(DSM\left(Concat\left(F_{fused}, F_{local}\right)\right)\right). \quad (12)$$

C. AECM

To effectively fuse the multiscale features rich in edge details output by PERA with deep semantic information, this article introduces the AECM. Through its core component, BAC-Net, the AECM replaces the upsampling and simple concatenation operations found in traditional FPN. By employing bidirectional attention and gating strategies, the framework facilitates reciprocal guidance and calibration, connecting low-level edge details with high-level semantic representations. Specifically, deep semantics guide the extraction of fine details, whereas shallow edges refine semantic representations. This synergistic calibration markedly enhances fusion quality. Furthermore, it facilitates the integration of a P2 high-resolution detection head, thereby boosting the sensitivity to tiny object detection.

1) BAC-Net: Fundamentally, BAC-Net is designed to synergistically fuse deep semantic context with shallow edge details, thereby enabling precise boundary refinement and accurate object localization. The structure of BAC-Net is illustrated in Fig. 5.

The process begins with a high-level input tensor $X_H \in \mathbb{R}^{C_H\times H_H\times W_H}$ and a low-level input tensor $X_L \in \mathbb{R}^{C_L\times H_L\times W_L}$. The detailed data processing workflow of BAC-Net is described as follows.

a) Feature projection and gating signal generation: Significant disparities exist between the input deep semantic features X_H and shallow edge features X_L regarding channel dimensions and semantic distributions. Consequently, direct feature interaction is prone to causing information misalignment and conflicts. Therefore, to facilitate interaction within

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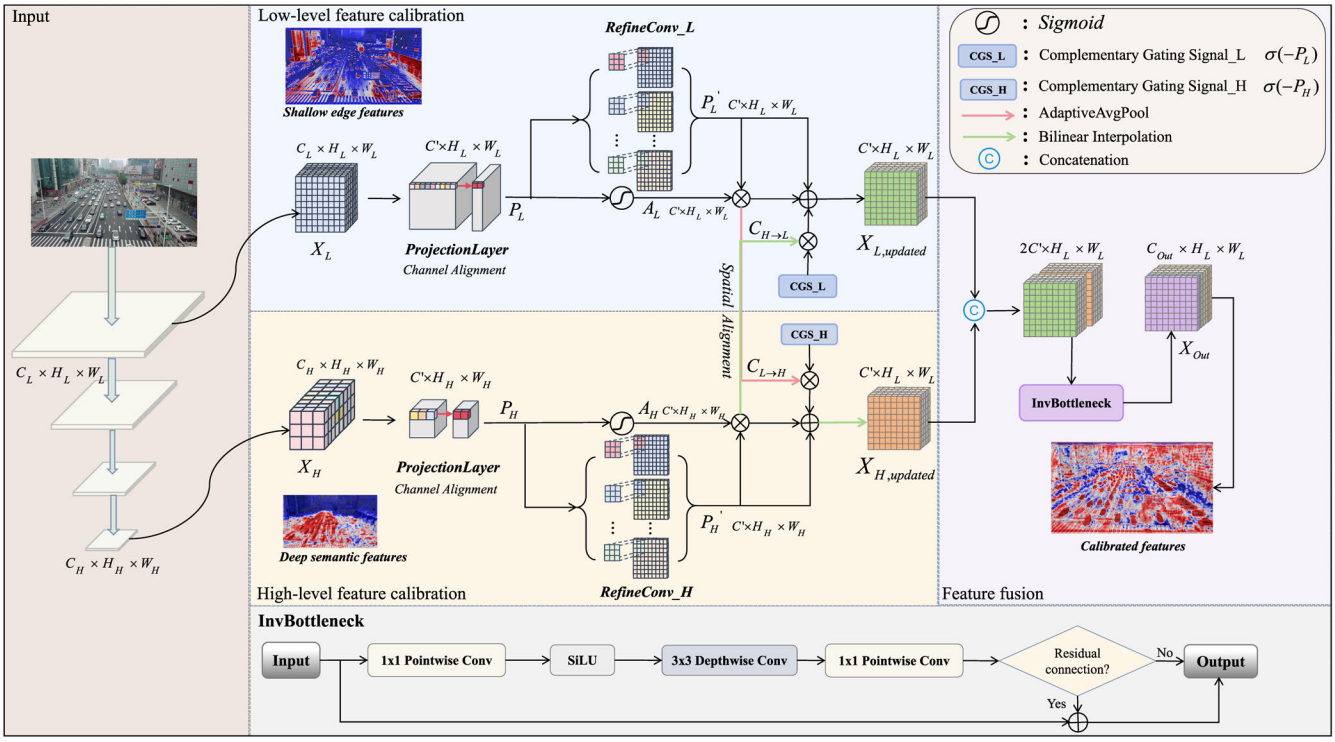


Fig. 5. Structure of the BAC-Net module. The *ProjectionLayer* performs channel alignment, while spatial alignment resolves resolution disparities between shallow and deep features prior to calibration.

a unified feature space, two independent 1×1 convolutional layers C_{proj} are employed. These layers linearly project X_L and X_H into a shared feature space with dimension C', yielding the projected features P_L and P_H

$$P_L = C_{\text{proj}}(X_L), \quad P_H = C_{\text{proj}}(X_H). \tag{13}$$

Simultaneously, to adaptively select effective information, P_L and P_H are processed by the sigmoid activation function $\sigma(\cdot)$. This generates soft spatial attention maps with values in the range of $[0, 1]$, which serve as gating signals A_L and A_H for subsequent information exchange. Subsequently, bidirectional feature calibration is performed using these gating signals

$$A_L = \sigma(P_L), \quad A_H = \sigma(P_H). \tag{14}$$

b) Bidirectional feature calibration: The projected features need further refinement. This process consists of two branches.

2) Calibration of Low-Level Features: The projected features P_L and P_H are individually processed by 3×3 depthwise convolutions (RefineConv_L, RefineConv_H). Prior to calibration, local spatial feature refinement is performed to generate P'_L and P'_H, to enrich them with local neighborhood information.

The update of the low-level feature P'_L aims to complement its context using high-level semantic information. First, the high-to-low context information $C_{H\to L}$ is generated. Specifically, gating signals are utilized to extract salient regions from high-level features ($A_H \odot P'_H$), which are then sampled to the low-level dimension via bilinear interpolation $\mathcal{F}_{bi}$

$$C_{H\to L} = \mathcal{F}_{\text{bi}}(A_H \odot P'_H). \tag{15}$$

Subsequently, the updated low-level feature $X_{L,updated}$ is obtained by summing three components

$$X_{L,\text{updated}} = P'_L + P'_L \odot A_L + \sigma(-P_L) \odot C_{H\to L}. \tag{16}$$

As shown in (16), the update of the low-level feature comprises three components: 1) the original refined feature P'_L serving as the basis; 2) the self-enhancement term $P'_L \odot A_L$, which reinforces intrinsic important edge details; and 3) the cross-layer calibration term $\sigma(-P_L) \odot C_{H\to L}$. By utilizing the product of the low-level complementary gating signal (CGS_L) and the high-level semantic information $C_{H\to L}$, this term selectively fills less significant regions. Consequently, the semantic representation is substantially augmented without compromising the original edge structure.

3) Calibration of High-Level Features: Symmetric to the calibration of low-level features, high-level features utilize precise edge details from the low level to achieve feature sharpening and spatial alignment. The process is described as follows:

$$C_{L\to H} = \text{AAP}(A_L \odot P'_L) \tag{17}$$

$$X_{H,\text{updated}} = P'_H + P'_H \odot A_H + \sigma(-P_H) \odot C_{L\to H}. \tag{18}$$

This step utilizes high-frequency edge information from low-level feature maps to calibrate and sharpen object boundaries within high-level semantic features, which are often blurred due to downsampling. Consequently, the localization precision of semantic features is improved.

Following bidirectional calibration, the updated high-level feature $X_{H,updated}$ is sampled to match the dimensions of $X_{L,updated}$ and concatenated along the channel dimension. To

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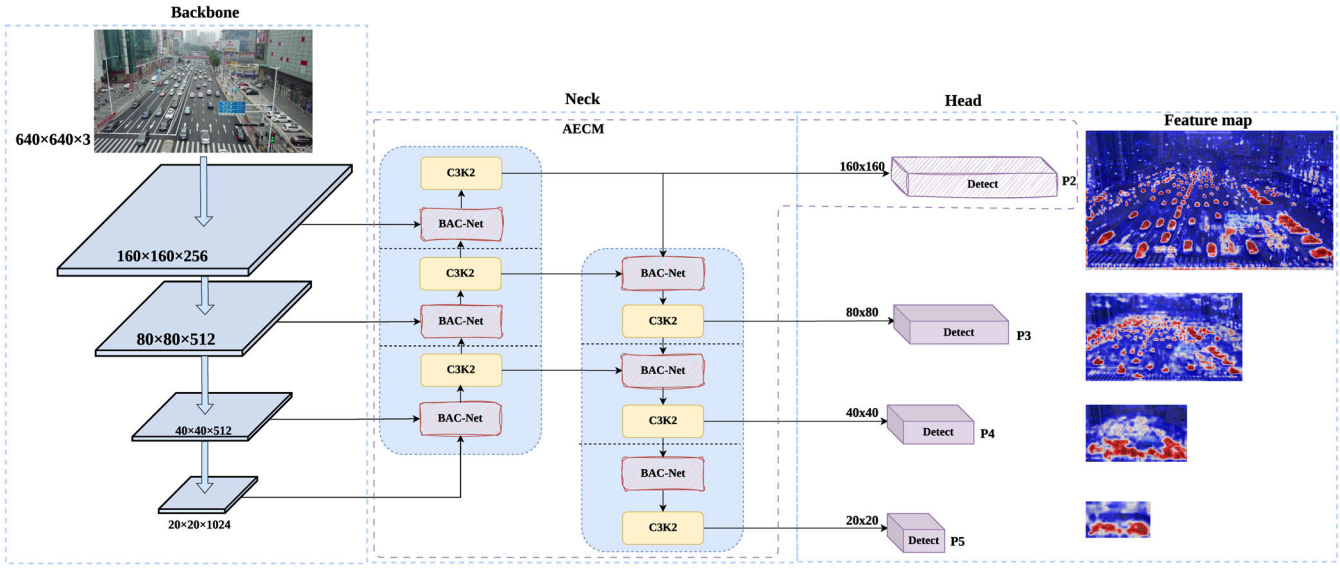


Fig. 6. Schematic of the AECM with the integrated P2 high-resolution detection path.

efficiently fuse these calibrated features, the computationally efficient IBN module [49] is adopted for channel and spatial fusion. Consequently, the final output feature X_{out} is obtained, possessing both deep semantic discriminability and shallow detail localization precision

$$X_{\text{out}} = \text{IBN}\left(\text{Concat}\left(\mathcal{F}_{\text{bi}}\left(X_{H,\text{updated}}\right), X_{L,\text{updated}}\right)\right) \tag{19}$$

where IBN(·) denotes the inverted bottleneck block.

4) P2 Detection Branch Within AECM: Traditional YOLO series models typically introduce the P2 detection layer through simple linear interpolation and channel concatenation within the FPN. While this approach increases high-resolution features, it often overlooks the semantic gap between deep semantics and shallow details. Furthermore, the input to traditional P2 heads consists of shallow backbone features, which usually contain substantial noise. In this study, the P2 detection head is seamlessly integrated into the AECM. Specifically, shallow features entering the P2 head first undergo refinement preprocessing by the PERA in the backbone to enhance high-frequency edge details and noise suppression. Then, in the neck, the BAC-Net utilizes deep semantics as a "gating signal" to perform bidirectional calibration on the spatial features of the P2 layer. As shown in Fig. 6, the calibrated features undergo further semantic refinement via the C3K2 module before being fed into the added P2 detection head. This design ensures that the P2 head receives semantic-rich features with clear edges, effectively suppressing background clutter that typically plagues high-resolution feature maps, thus improving the overall detection robustness and accuracy.

D. Adaptive Weight Focaler-MPDIoU Loss Function

UAV small object detection faces dual challenges: sample imbalance and localization sensitivity. Although existing ranking-based or distribution-aware loss functions perform excellently in general domains, complex distribution modeling is prone to introducing background noise in low-pixel UAV scenarios, and ranking mechanisms struggle to provide robust gradient guidance. To address this, this article proposes an adaptive weight Focaler-MPDIoU loss function that balances dynamic equilibrium and geometric sensitivity. This function alleviates sample distribution imbalance through the Focaler-IoU mechanism [15] and utilizes MPDIoU [16] to reinforce vertex geometric constraints for improved localization accuracy. An adaptive weight α is introduced, enabling the model to achieve a smooth transition from coarse localization to fine-grained tuning based on the training state. Without increasing the computational overhead, this strategy significantly enhances the robustness and convergence efficiency of small object detection training, achieving a more stable sample balancing effect than distribution-aware methods.

1) Focaler-IoU Mechanism Addressing Sample Imbalance: To mitigate the imbalance between hard and easy samples in bounding box regression (BBR), the Focaler-IoU mechanism [15] is introduced. By reconstructing IoU values through linear interval mapping, this mechanism dynamically adjusts loss weights for samples of different qualities. The formula for Focaler-IoU [15] is expressed as follows:

$$\text{IoU}_{\text{Focaler}} = \begin{cases} 0, & \text{IoU} < d \\ \dfrac{\text{IoU} - d}{u - d}, & d \le \text{IoU} \le u \\ 1, & \text{IoU} > u. \end{cases} \tag{20}$$

Here, d and u are thresholds defining the interval of focused samples. The mechanism suppresses the gradient contribution of high-quality samples ($IoU > u$) to avoid overfitting on well-optimized examples. Simultaneously, it provides significant optimization signals for medium-quality samples ($d \le IoU \le u$). This directs model training toward prediction boxes with improvement potential. The loss function $\mathcal{L}_{Focaler}$ is defined as follows:

$$\mathcal{L}_{\text{Focaler}} = 1 - \text{IoU}_{\text{Focaler}}. \tag{21}$$

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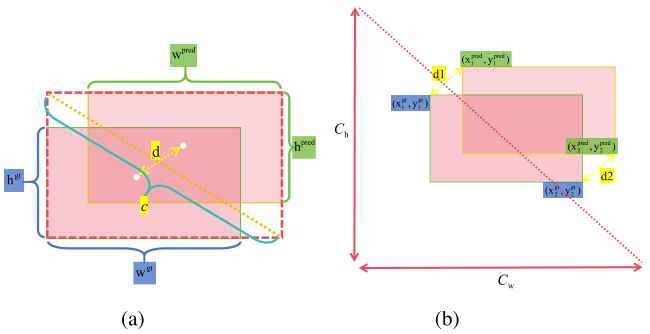


Fig. 7. Schematic comparison between (a) CIoU and (b) MPDIoU. The green and yellow boxes represent the ground truth and predicted boxes, respectively. In (a), c denotes the diagonal length of the smallest enclosing box. In (b), C_w and C_h represent the width and height of the input image, respectively.

In our implementation, d is fixed at 0 to ensure that all focused samples contribute effective gradients. However, this mechanism primarily optimizes sample weights, and its capability to enhance the geometric localization accuracy is limited.

2) Superior Geometric Supervision via MPDIoU Loss: CIoU [50] functions as an indirect supervision mechanism. As illustrated in Fig. 7(a), it optimizes bounding boxes by penalizing the central point distance and the aspect ratio difference between the predicted and ground-truth boxes (calculated via their respective widths and heights w^{gt}, h^{gt} and w^{pred}, h^{pred}). For small objects, central points may be in very close proximity, leading to diminished gradient signals. Simultaneously, the aspect ratio serves as a relatively indirect constraint, leading to slow model convergence. To enhance the geometric supervision accuracy of BBR, MPDIoU [16] is adopted, as shown in Fig. 7(b). By minimizing the Euclidean distances between the top-left and bottom-right vertices of the predicted and ground truth boxes, MPDIoU [16] provides a precise and efficient geometric metric for the minute displacements of small objects. Consequently, it offers stronger convergence momentum for fine localization. The distances d_1 and d_2 between the two vertices are calculated as follows:

$$d_1^2 = \left(x_1^{\text{pred}} - x_1^{\text{gt}}\right)^2 + \left(y_1^{\text{pred}} - y_1^{\text{gt}}\right)^2 \tag{22}$$

$$d_2^2 = \left(x_2^{\text{pred}} - x_2^{\text{gt}}\right)^2 + \left(y_2^{\text{pred}} - y_2^{\text{gt}}\right)^2. \tag{23}$$

MPDIoU Loss [16] integrates the IoU metric with a vertex distance penalty term, defined as follows:

$$\mathcal{L}_{\text{MPDIoU}} = 1 - \text{IoU} + \frac{d_1^2}{C_w^2 + C_h^2} + \frac{d_2^2}{C_w^2 + C_h^2}. \tag{24}$$

MPDIoU Loss [16] is highly sensitive to minute displacements of small objects, enabling it to provide finer gradient guidance and thereby significantly improve the localization accuracy. However, it does not inherently address the issue of sample imbalance.

3) Dynamic Weight Fusion Mechanism: Static weighted fusion fails to adapt to the dynamic requirements of the model across different training stages. To address this, a dynamic weight scheduling mechanism driven by the model's intrinsic localization accuracy is formulated. This mechanism utilizes the average IoU ($\overline{\text{IoU}}$) within each training batch as a training progress indicator to dynamically regulate the fusion weight α between Focaler-IoU [15] and MPDIoU Loss [16]

$$\alpha = \left(\overline{\text{IoU}}\right)^{\beta}. \tag{25}$$

Here, $\overline{\text{IoU}}$ represents the average intersection over union of all positive samples within the current training batch, and β is a hyperparameter controlling the rate of weight variation. The final adaptive loss function $\mathcal{L}_{\text{Adaptive}}$ is defined as follows:

$$\mathcal{L}_{\text{Adaptive}} = \alpha \cdot (1 - \text{IoU}_{\text{Focaler}}) + (1 - \alpha) \cdot \left[1 - \text{IoU} + \frac{d_1^2 + d_2^2}{C_w^2 + C_h^2 + \epsilon}\right] \tag{26}$$

where ϵ is a small positive number ensuring numerical stability.

The distinct advantage of this design lies in its adaptive optimization process, which endows the loss function with the capability to perceive the training state and perform self-adjustment. This mechanism imposes appropriate supervision signals according to the model's learning progress, enabling the loss function to prioritize MPDIoU Loss [16] during the early training stage (low $\overline{\text{IoU}}$) to rapidly acquire geometric localization and smoothly transition to prioritizing Focaler-IoU [15] in the later stage (high $\overline{\text{IoU}}$) to fine-tune the boundaries of focused samples. This facilitates a smooth transition from coarse localization to fine-grained tuning, thereby efficiently enhancing detection performance.

IV. Experiments

A. Experimental Setup

1) Datasets: To validate the performance of the proposed AEC-YOLO, the publicly available UAV detection benchmark, the VisDrone2019 [17], AI-TOD [18], DOTA-v1.5 [19], and DIOR [20] datasets are selected.

The VisDrone2019 dataset [17] was constructed by the AISKYEYE team at Tianjin University. This dataset is officially partitioned into 6471 for training, 548 for validation, and 1610 for testing. The dataset encompasses ten categories of typical objects from UAV perspectives, with data acquisition scenarios spanning urban and rural environments. Characterized by challenging features—including drastic fluctuations in object density, complex backgrounds, and a massive preponderance of small-scale objects—it is widely recognized as a premier benchmark for UAV small object detection.

AI-TOD [18] is a dataset specifically designed for tiny object detection in remote sensing imagery, containing eight categories with over 7 00 000 object instances and an average object size of 12.8 pixels. Following the official split, 11 214 images are used for training, 2804 for validation, and 14 018 for testing.

As an upgraded version of the DOTA series, the DOTA-v1.5 [19] dataset incorporates a larger number of extremely tiny object instances, covering 16 categories and containing 4 03 318 object instances. Compared with the previous version, v1.5 significantly increases detection difficulty due to more complex backgrounds, denser object arrangements, and high-frequency noise in high-resolution aerial

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(a) (b) (c) (d)

Fig. 8. Representative samples randomly selected from four benchmark datasets. (a) VisDrone2019 (UAV-based high-density small objects), (b) AI-TOD (extremely tiny objects with low resolution), (c) DOTA-v1.5 (multiscale and multicategory targets), and (d) DIOR (large-scale with diverse geographical backgrounds).

images. This study utilizes the DOTA-v1.5 dataset to verify the model's adaptability to diverse objects.

The DIOR dataset [20] is one of the most representative large-scale benchmark datasets in the Earth observation field. It contains 23 463 images and 1 92 472 object instances across 20 common categories. The dataset exhibits high interclass similarity and intraclass diversity, with object scales spanning a vast range. Evaluating AEC-YOLO on the DIOR dataset aims to demonstrate the model's generalization capability across extensive geographical environments and object distributions.

Representative data samples are depicted in Fig. 8.

2) Experimental Environment and Training Parameters: To ensure the fairness of experimental comparisons, a unified hyperparameter configuration was adopted for all training and validation processes. All experiments in this study were conducted under the same hardware and software environment: the operating system was Ubuntu 20.04, and the deep learning framework was PyTorch 1.11.0 with CUDA 11.3 acceleration. Training and testing were executed on a single NVIDIA GeForce RTX 3090 GPU (24-GB Video Random Access Memory, VRAM). The training input image resolution was 640×640, the batch size was set to 16, and the number of epochs was 300. Training utilized a cosine annealing learning rate scheduler and the stochastic gradient descent (SGD) optimizer (learning rate $lr = 0.01$, weight decay = 0.0005, and momentum = 0.937), with mosaic data augmentation enabled. All experiments for the baseline models and the proposed method were based on the Ultralytics framework (v8.3.9). To mitigate the impact of randomness during training, three independent trials were performed for key experiments (ablation and comparative experiments on the VisDrone2019 dataset), with each trial using a different random seed. Final results are reported in the form of mean $\pm$ standard deviation.

3) Evaluation Metrics: To quantitatively evaluate the performance of the proposed AEC-YOLO model, standard evaluation metrics in the field of object detection are adopted. This holistic assessment encompasses detection accuracy, recall rates, localization precision, model complexity, and inference efficiency.

Precision (P) and Recall (R): These metrics quantify the accuracy of predictions and the capability to detect ground truth objects, respectively. They are derived based on true positives (TPs), false positives (FPs), and false negatives (FNs). The calculation formulas are as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{27}$$

Mean Average Precision (mAP): Serving as a pivotal metric, mAP evaluates the overall performance of the model across various confidence thresholds by calculating the area under the precision–recall (P–R) curve. Average precision (AP) denotes the area under the P–R curve for a specific class, whereas mAP denotes the mean AP across all N categories. The mathematical formulations are as follows:

$$\mathrm{AP} = \int_0^1 P(R)\, dR \tag{28}$$

$$\mathrm{mAP} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{AP}_i. \tag{29}$$

This study primarily utilizes the following two mAP specifications.

mAP_{50}: The mAP calculated at a fixed intersection over union (IoU) threshold of 0.5. It measures the model's capability for object recognition and localization tolerance.

$mAP_{50:95}$: The standard metric for the COCO [51] challenge. It calculates the average of ten mAP values over IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05. This metric imposes stricter requirements on localization accuracy; a higher value indicates superior BBR precision.

The $F1$-score, defined as the harmonic mean of precision and recall, offers a balanced evaluation of the classification and localization performance of the object detection model regarding positive samples. The calculation formula is as follows:

$$\mathrm{F1} = \frac{2 \times P \times R}{P + R}. \tag{30}$$

In addition, GFLOPs and Params are utilized to evaluate model complexity. Inference efficiency is measured using frames per second (FPS) and latency (total time required to process a single image), both of which are measured during single-image inference ($Batch\ Size = 1$).

B. Ablation Experiments

This section first verifies the independent effectiveness and synergy of each core component in the AEC-YOLO

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TABLE II
ABLATION EXPERIMENTS ON THE VISDRONE2019 DATASET

Baseline	PERA	BAC-Net	P2	$\mathcal{L}_{\text{Adaptive}}$	Precision (%)	Recall (%)	mAP_{50} (%)	$\text{mAP}_{50:95}$ (%)	Params (M)	GFLOPs	FPS	Latency (ms)
✓					49.42 ± 0.25	38.63 ± 0.31	38.83 ± 0.12	23.41 ± 0.10	9.42	21.3	136	7.35
✓	✓				52.34 ± 0.12	39.21 ± 0.23	40.46 ± 0.25	24.48 ± 0.21	9.8	28.7	112	8.93
✓		✓			54.41 ± 0.21	41.53 ± 0.19	43.51 ± 0.26	27.21 ± 0.23	12.7	35.9	96	10.42
✓			✓		52.12 ± 0.15	40.21 ± 0.23	41.81 ± 0.33	25.33 ± 0.13	9.56	30.4	99	10.1
✓				✓	52.83 ± 0.12	38.88 ± 0.15	40.16 ± 0.18	23.64 ± 0.08	9.41	21.3	134	7.46
✓		✓	✓		55.71 ± 0.12	44.38 ± 0.22	46.55 ± 0.16	28.73 ± 0.15	13.1	50.3	68	14.7
✓	✓		✓		53.83 ± 0.13	43.11 ± 0.16	44.53 ± 0.17	26.83 ± 0.12	9.94	37.8	82	12.2
✓	✓	✓			54.84 ± 0.18	42.12 ± 0.13	44.21 ± 0.12	27.42 ± 0.15	13.2	37.3	86	11.6
✓	✓	✓	✓		56.60 ± 0.23	45.71 ± 0.23	47.58 ± 0.11	29.56 ± 0.15	13.6	51.7	63	15.9
✓	✓	✓	✓	✓	**58.13 ± 0.15**	**45.93 ± 0.11**	**48.26 ± 0.21**	**29.86 ± 0.16**	13.6	51.7	61	16.39

The symbol "✓" denotes the integration of the corresponding proposed component. The best results in accuracy metrics are highlighted in **bold**.

framework. Subsequently, it will explore the effectiveness analysis of the PERA multiscale combination and the adaptive weight loss function.

1) Analysis of Core Component Effectiveness and Module Redundancy: To systematically verify the effectiveness of each core component in the AEC-YOLO proposed framework, we conducted a series of detailed ablation experiments. Using YOLO11s as the baseline model, the proposed PERA, BAC-Net, P2 layer, and adaptive weight Focaler-MPDIoU loss function were progressively integrated. All experiments were performed under identical conditions, with three independent trials conducted for each configuration. Final results are reported as mean ± standard deviation, as presented in Table II.

The precision of the baseline YOLO11s model is 49.42%, and the mAP_{50} is 38.83%. When PERA is individually integrated into the backbone network, the precision of the model increases by 2.92%, and the mAP_{50} improves to 40.46% (+1.63%), with Params slightly increasing by only 0.38M. This result strongly demonstrates the effectiveness of PERA, which significantly enhances feature quality with an extremely low parameter increment.

After independently integrating BAC-Net into the neck network, the mAP_{50} of the model significantly increases to 43.51% (+4.68%), and the $\text{mAP}_{50:95}$ reaches 27.21% (+3.8%). Although Params increase to 12.7 M and GFLOPs rise to 35.9, BAC-Net achieves nonlinear calibration complementarity between deep semantics and shallow edge details. It effectively resolves the misalignment and dilution issues caused by traditional FPN feature concatenation, making it the component with the most significant performance gain among single modules.

By adopting the P2 high-resolution detection head, precision improves to 52.12% (+2.7%), and mAP_{50} increases to 41.81% (+2.98%). Due to the massive number of bounding boxes processed by the P2 layer, the FPS drops to 99, and the latency increases to 10.1 ms. This result indicates that high-resolution features are crucial for tiny object perception, and their computational overhead remains within an acceptable range for real-time processing.

Furthermore, when the adaptive weight Focaler-MPDIoU loss function is validated individually, it improves precision by 3.41% without increasing Params or GFLOPs, with only a negligible change in inference speed. By balancing hard sample weights and optimizing localization supervision signals, this loss function improves detection accuracy at the training objective level.

Further experiments show that after combining BAC-Net and the P2 detection head to construct the AECM, the mAP_{50} increases to 46.55% (+7.72%), showing a complementary trend. This significant improvement demonstrates that our P2 design is not a redundant scale expansion. Under the synergistic interaction of PERA and the AECM, the mAP_{50} reaches 47.58% (+8.75%). This complementary drive of edge refining in the feature extraction stage and cross-layer calibration in the feature fusion stage achieves a stepwise growth in model performance.

From the perspective of componentwise removal analysis, PERA and AECM (composed of BAC-Net and the P2 detection head) are not a functionally redundant module stack. Specifically, using the complete synergistic combination (PERA + AECM) as the reference, removing PERA (i.e., using only the AECM) decreases the mAP_{50} to 46.55% (−1.03%). This indicates that although the AECM possesses certain cross-layer calibration capabilities, it still requires the source edge enhancement provided by PERA in the backbone network. When BAC-Net is removed (i.e., adopting the PERA + P2 combination), the mAP_{50} sharply drops to 44.53% (−3.05%). This severe degradation demonstrates that without the cross-layer bidirectional calibration, the edge information extracted by PERA struggles to align fully with deep semantics. Directly feeding uncalibrated features into the P2 detection head leads to performance impairment. When the P2 detection head is removed (i.e., adopting the PERA + BAC-Net combination), the mAP_{50} decreases to 44.21% (−3.37%). This occurs because edge enhancement and semantic calibration alone are insufficient to fully activate the detection responses for tiny objects. Therefore, the high-resolution detection branch plays a necessary role in improving small object recall. Furthermore, if the entire AECM is removed and only PERA is retained, the mAP_{50} falls to 40.46% (−7.12%). This highlights the critical role of the AECM in edge-semantic fusion and detection response transformation. In summary, these modules operate at three distinct stages:

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TABLE III
EXPERIMENTAL RESULTS OF PERA MULTISCALE COMBINATIONS

Scale	Precision (%)	Recall (%)	mAP_{50} (%)	$mAP_{50:95}$ (%)	Params (M)
[3, 6, 9, 12, 15]	51.12	39.14	40.13	23.91	9.92
[2, 4, 6, 9, 12]	49.91	39.22	39.74	23.93	9.91
[3, 6, 9, 12]	52.34	39.21	40.46	24.48	9.80

feature generation, feature calibration, and detection response. They form a complete logical closed loop without functional redundancy.

Ultimately, the integrated AEC-YOLO achieves the optimal mAP_{50} of 48.26%, which is a significant improvement of 9.43 percentage points over the baseline. The $mAP_{50:95}$ also increases by 6.45 percentage points. Furthermore, it is observed that the standard deviations across all configurations remain within a narrow range, significantly smaller than the performance gains introduced by each module. This confirms that the performance gains of AEC-YOLO stem from structural optimization rather than random training fluctuations, indicating that the proposed model achieves a balanced performance improvement in small object detection.

Although the number of parameters increases from 9.42M to 13.60M, and the GFLOPs reach 51.7, studies on real-time performance indicate that the real-time requirement for UAV object detection is typically defined as 30–60 FPS, maintaining an inference latency of <50 ms is considered sufficient to meet real-time inference demands [52], [53]. Limited by the physical sampling frequency of airborne sensors, excessively high inference frame rates often manifest as unutilized computational redundancy. While maintaining a real-time inference speed of 61 FPS, AEC-YOLO achieves a significant leap in detection accuracy by converting the surplus frame rates that exceed real-time requirements, while concurrently preserving low latency. This demonstrates an exceptional accuracy-efficiency tradeoff capability. In summary, the ablation experiments fully validate the independent contributions and synergistic gains of PERA, the AECM, and the adaptive loss function, systematically verifying the effectiveness of the proposed framework.

2) Experiments on PERA Multiscale Combination Effectiveness: The pivotal attribute of PERA resides in its capacity to capture context and edge information at varying scales through multiple parallel branches. To validate the rationale of the selected multiscale combination and investigate the impact of different scale ranges on model performance, comparative experiments targeting the scale parameters within PERA were designed. Using the four-scale combination $[3, 6, 9, 12]$ as the baseline, two additional five-scale combinations were explored: $[3, 6, 9, 12, 15]$ (investigating coarser-grained receptive fields) and $[2, 4, 6, 9, 12]$ (investigating finer-grained receptive fields). The experimental results are presented in Table III.

Experimental results indicate that the four-branch combination $[3, 6, 9, 12]$ achieved an mAP_{50} of 40.46% and an $mAP_{50:95}$ of 24.48%, identifying it as the optimal configuration. Incorporating larger scales leads to performance degradation caused by the aggregation of excessive background noise, whereas introducing smaller scales results in deterioration due to an over-emphasis on local details and a lack of contextual understanding. This combination strikes the optimal balance between fine-grained edge capture and contextual information aggregation, thereby validating its effectiveness.

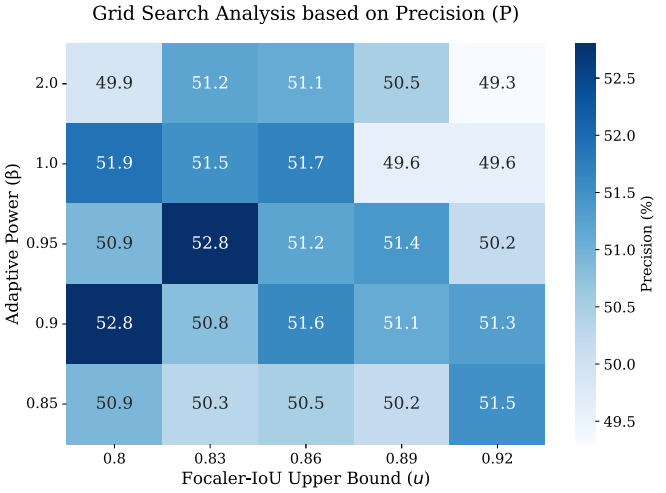


Fig. 9. Grid search experimental results for hyperparameters β and u of the adaptive weight Focaler-MPDIoU loss.

3) Effectiveness Analysis of Adaptive Weight Loss Function:

a) Analysis of hyperparameters β and u: In the analysis of hyperparameters for the adaptive weight loss function, a grid search experiment was conducted to scrutinize the impact of the adaptive power parameter β and the Focaler-IoU [15] upper bound u on model performance. The experimental results are presented in Fig. 9. To prioritize optimizing the model's anti-interference capability in complex backgrounds and to evaluate its ability to suppress FPs, precision (P) was selected as the primary evaluation metric. All experimental results represent the final performance metrics evaluated after completing a full 300 epochs of training.

Experimental results reveal a coupling effect between β and u, indicating that performance peaks are not isolated occurrences. Specifically, two distinct parameter configurations, ($\beta = 0.9, u = 0.8$) and ($\beta = 0.95, u = 0.83$), concurrently attained a maximum precision of 52.80%. This confirms the necessity of jointly optimizing these parameters as a coupled system. Furthermore, the resulting heatmap indicates that the sensitivity of the model to β and u is moderately controlled. Within the optimal neighborhood ($\beta \in [0.9, 1.0]$, $u \in [0.8, 0.86]$), precision varies smoothly, forming

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TABLE IV
EXPERIMENTAL COMPARISON OF DIFFERENT LOSS FUNCTIONS ON THE VISDRONE2019 DATASET

Loss Function	P	R	mAP_{50}	$mAP_{50:95}$
YOLO11s CIoU (Baseline)	49.42	38.63	38.83	23.41
MPDIoU Loss	51.61	37.72	39.14	23.53
Focaler-IoU Loss	50.62	38.81	39.63	**23.72**
Focaler-MPDIoU Loss	51.13	38.42	39.51	23.61
GFL Loss	45.52	35.94	36.12	22.33
RS Loss	51.34	38.31	39.82	23.32
$\mathcal{L}_{\text{Adaptive}}$ (Ours)	**52.83**	**38.88**	**40.16**	23.64

Note: P, R, and mAP values are reported in percentage (%). P and R denote Precision and Recall, respectively. Bold values indicate the best results.

a performance plateau. Even under the most unfavorable parameter combination, precision remains ≥49.3%, which approaches the baseline level. This confirms the robustness and parameter fault tolerance of AEC-YOLO.

Furthermore, in order to determine the unique best configuration from these two candidates with parallel performance, $mAP_{50:95}$ was introduced as the decisive supplementary metric, as it provides a more comprehensive reflection of the model's overall detection performance. The $mAP_{50:95}$ values corresponding to the two parameter sets were 23.6% and 23.4%, respectively. Consequently, ($\beta = 0.9, u = 0.8$) was selected as the optimal hyperparameter combination. By facilitating a rapid transition (via a suitable β value) to a fine-tuning stage with a relatively lenient target (via a low u value), this combination encourages the model to adopt a more cautious prediction strategy, robustly mitigating FPs and maximizing precision.

b) Effectiveness analysis of $\mathcal{L}_{\text{adaptive}}$: To verify the effectiveness of $\mathcal{L}_{\text{Adaptive}}$, comprehensive comparisons were conducted against the baseline loss function, single components, fixed-weight combinations, as well as advanced distribution-aware loss (GFL Loss) [36] and ranking loss (RS Loss) [35]. The experimental results are presented in Table IV.

The results demonstrate that the individual components of the loss function exhibit distinct strengths: MPDIoU Loss [16] significantly enhances localization precision (P) albeit at the expense of recall (R); conversely, Focaler-IoU Loss [15] excels in addressing sample imbalance, yielding a more notable improvement in mAP_{50}. However, a simple combination of the two using fixed weights results in performance across multiple metrics that is inferior to using Focaler-IoU Loss [15] alone, thereby exposing the limitations of static balancing strategies.

Although the distribution-aware loss (GFL Loss) [36] performs excellently in general detection, its performance degrades significantly in UAV scenarios (mAP_{50} is only 36.12%, and precision drops to 45.52%). This is because extremely small objects in UAV images suffer from pixel scarcity and texture absence; therefore, forcibly fitting complex probability distributions easily introduces background noise. The ranking-based loss (RS Loss) [35] mitigates foreground–background imbalance by optimizing sample ranks, demonstrating excellent performance (precision reaches 51.34% and mAP_{50} reaches 39.82%). However, under massive background interference, its vast ranking search space restricts the geometric sensitivity to the minute displacements of extremely small objects.

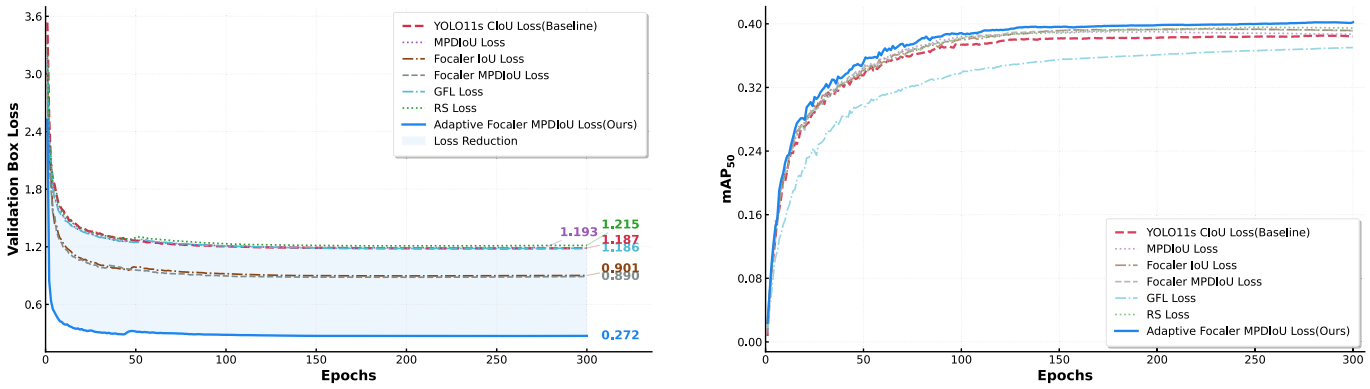
In contrast, our proposed $\mathcal{L}_{\text{Adaptive}}$ achieves competitive performance. While maintaining a high recall, it increases precision to an optimal 52.83%, an improvement of 3.41 percentage points over the baseline, and achieves an mAP_{50} of 40.16%. The convergence curves of various loss functions and the mAP_{50} curves during the validation phase, as illustrated in Fig. 10, further corroborate this advantage. Fig. 10(a) demonstrates that the proposed $\mathcal{L}_{\text{Adaptive}}$ (blue curve) converges fastest during the initial training stage, with the final loss stabilizing at 0.272, significantly lower than the baseline and methods such as GFL Loss [36] and RS Loss [35]. The mAP_{50} comparison curve in Fig. 10(b) indicates that the proposed method (blue curve) achieves the highest accuracy upper bound (40.16%) throughout the entire training cycle. This result fully demonstrates the superiority of the proposed method in enhancing convergence efficiency and localization robustness without introducing any additional computational parameters. By flexibly adjusting the weight contributions of Focaler-IoU [15] and MPDIoU Loss [16], Although Focaler-IoU obtains a slightly higher $mAP_{50:95}$, the proposed $\mathcal{L}_{\text{Adaptive}}$ achieves the best precision, recall, and mAP_{50}, showing a more balanced overall performance.

C. Comparative Experiments

1) Comparison With the Baseline YOLO11s Model: To validate the effectiveness of our framework, we compare AEC-YOLO with the baseline model YOLO11s on the VisDrone2019 [17] dataset. As shown in Fig. 11, AEC-YOLO outperforms the baseline in terms of mAP_{50} across all ten categories. Further category-level analysis reveals the underlying mechanism of our edge enhancement strategy. The model achieves the most significant gains on typical small objects with clear high-frequency boundaries and strong geometric priors, such as "people" (+13.5%), "pedestrian" (+13.3%), and "motor" (+12.8%). This is primarily attributed to the PERA module, which actively captures and sharpens the rigid contours of these objects. Furthermore, the AECM effectively preserves key boundary cues during multiscale feature alignment. Together, these modules significantly suppress background interference in UAV views. In contrast, categories such as "awning-tricycle" (+5.7%) and "car" (+5.9%, which already has a high baseline) show limited improvement. This is because these objects often exhibit irregular shapes or suffer from severe boundary blending with the environment, thus lacking consistent high-frequency physical contours. When the intrinsic edge signals of the targets are degraded or naturally ambiguous, the performance gains from edge enhancement naturally diminish. This distribution of performance gains, which strongly aligns with the geometric properties of the objects, further validates that the active protection and calibration of edge information are the core drivers behind our framework's performance improvements.

Second, judging from the comprehensive performance curves, AEC-YOLO exhibits significant performance superiority. Figs. 12 and 13 illustrate the experimental results of YOLO11s and AEC-YOLO on the VisDrone2019 [17] dataset.

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(a) (b)

Fig. 10. Comparison of training convergence and performance using different loss functions on the VisDrone2019 dataset. (a) Convergence curves of the BBR loss during the validation phase. (b) mAP_{50} performance comparison curves.

TABLE V
PERFORMANCE COMPARISON OF AEC-YOLO WITH OTHER SOTA OBJECT DETECTION MODELS ON THE VISDRONE2019 DATASET

Model	Precision (%)	Recall (%)	mAP_{50} (%)	$mAP_{50:95}$ (%)	Params (M)	GFLOPs	FPS
RT-DETR†	39.58 ± 0.35	35.79 ± 0.22	33.97 ± 0.18	19.46 ± 0.13	20.1	61.2	46
Faster R-CNN†	37.23 ± 0.11	26.96 ± 0.21	30.68 ± 0.19	17.76 ± 0.15	41.38	134	23
Cascade R-CNN†	36.51 ± 0.13	25.88 ± 0.23	29.57 ± 0.12	17.25 ± 0.16	69.17	162	18
QueryDet [54]	-	39.3	48.1	28.3	39.48	226.19	37.52
CEASC [55]	-	35.1	43.4	24.6	-	132.91	21.71
FBRT-YOLO [56]	-	-	42.4	25.9	2.9	22.9	143
YOLOv3-tiny†	38.96 ± 0.34	23.81 ± 0.28	23.31 ± 0.55	13.03 ± 0.31	12.13	18.93	142
YOLOv5s†	51.15 ± 0.32	37.42 ± 0.25	39.12 ± 0.28	23.41 ± 0.1	9.12	23.85	118
YOLOv6s†	49.63 ± 0.25	36.21 ± 0.22	37.95 ± 0.32	22.74 ± 0.26	16.30	44.04	68
YOLOv8s†	50.94 ± 0.2	37.85 ± 0.29	39.22 ± 0.15	23.58 ± 0.22	11.13	28.46	105
YOLOv9s†	52.35 ± 0.32	38.51 ± 0.18	39.84 ± 0.24	23.98 ± 0.10	7.17	26.75	112
YOLOv10s†	50.28 ± 0.16	37.72 ± 0.20	39.48 ± 0.11	23.61 ± 0.12	7.22	21.43	140
YOLO11s†	49.42 ± 0.25	38.63 ± 0.31	38.83 ± 0.12	23.41 ± 0.10	9.42	21.3	136
YOLOv12s†	49.42 ± 0.28	38.64 ± 0.32	39.51 ± 0.25	23.78 ± 0.24	9.23	21.23	138
YOLOv13s†	48.81 ± 0.25	37.28 ± 0.19	37.58 ± 0.22	22.25 ± 0.21	9	20.68	142
EWS-YOLO [57]	-	37.8	42.4	24.1	6.5	22.8	332
EMFE-YOLO [58]	55.2	45.3	46.9	29.1	3	33.1	121
HSF-YOLO [59]	51.6	40	41	24.6	3.4	17.1	-
YOLO-MARS [60]	50.2	40	40.9	23.4	2.93	-	-
TA-YOLO-s [61]	53.9	44.3	45.4	27.7	13.9	43.3	-
MEP-YOLOv5s [62]	55	45.9	44.5	25.9	22.5	112.4	-
AEC-YOLO (Ours)†	**58.13 ± 0.15**	**45.93 ± 0.11**	**48.26 ± 0.21**	**29.86 ± 0.16**	13.6	51.7	61

† indicates models implemented in our local environment following a uniform experimental setup. These models were executed for three independent experiments, and the standard deviation range is presented in the table. Other comparative models, limited by environmental configurations or code accessibility, directly cite the optimal performance reported in their original publications to ensure a fair comparison. The best results in accuracy metrics are highlighted in **bold**.

The comparison of P–R curves reveals that the curve of AEC-YOLO lies above and envelops that of the baseline model across the majority of the range. This demonstrates the proposed method's superior robustness across various confidence levels. Simultaneously, the $F1$-confidence curves indicate that the optimal $F1$-score of the proposed model improved from 0.43 to 0.50, representing a superior balance achieved between precision and recall.

2) Comparison With Other Object Detection Models: To comprehensively evaluate the performance of the proposed AEC-YOLO model, extensive quantitative comparisons were conducted against current mainstream object detectors on the VisDrone2019 [17] dataset. The comparative models encompass classic frameworks such as Faster R-CNN [8] and Cascade R-CNN [63]; the Transformer-based RT-DETR [64]; several SOTA models (QueryDet [54], CEASC [55], and FBRT-YOLO [56]); the standard YOLO series (v3-tiny [24] and v5s-v13s [25], [26], [28], [29], [30], [31], [32], [33]); and other advanced models optimized for UAV scenarios (EWS-YOLO [57], EMFE-YOLO [58], HSF-YOLO [59], YOLO-MARS [60], TA-YOLO-s [61], and MEP-YOLOv5s [62]). The experimental results are presented in Table V.

The experimental results clearly indicate that AEC-YOLO exhibits a significant advantage in detection accuracy.

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TABLE VI
SUMMARY OF KEY TRAINING CONFIGURATIONS AND INFERENCE STRATEGIES OF THE COMPARED ADVANCED MODELS

Model	Input Res.	Epochs	Augmentation Strategy	Pre-training	Slicing Strategy
QueryDet [54]	1600×1600	124	Random multi-scale scaling, horizontal flip	ImageNet pre-training	Yes
CEASC [55]	1333×800	15	Random multi-scale scaling, horizontal flip	ImageNet pre-training	No
FBRT-YOLO [56]	640×640	300	Mosaic data augmentation	–	No
EWS-YOLO [57]	640×640	200	Mosaic data augmentation	No (from scratch)	No
EMFE-YOLO [58]	640×640	300	Mosaic data augmentation	–	No
HSF-YOLO [59]	640×640	200	Mosaic data augmentation	–	No
YOLO-MARS [60]	640×640	200	Mosaic data augmentation	No (from scratch)	No
TA-YOLO-s [61]	640×640	500	Mosaic data augmentation	No (from scratch)	No
MEP-YOLOv5s [62]	640×640	300	Mosaic data augmentation	No (from scratch)	No
AEC-YOLO (Ours)	640×640	300	Mosaic data augmentation	No (from scratch)	No

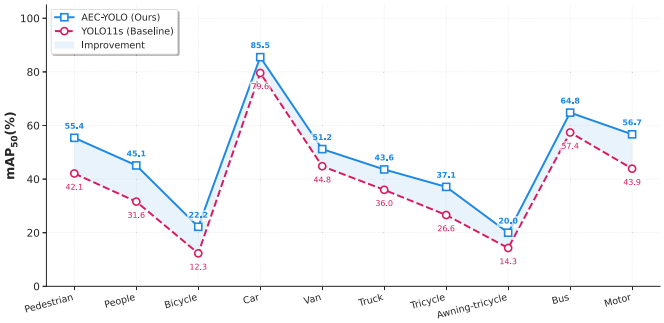


Fig. 11. Comparison of categorywise mAP_{50} performance between the baseline model and AEC-YOLO on the VisDrone2019 dataset.

Its mAP_{50} reaches 48.26%. Compared with the baseline YOLO11s model, the mAP_{50} of AEC-YOLO improves by 9.43 percentage points, significantly outperforming the latest models such as YOLOv13s (37.58%). Even when compared with advanced models specifically optimized for UAVs, such as EMFE-YOLO (46.9% mAP_{50}), AEC-YOLO still maintains a leading position. Furthermore, the standard deviations of AEC-YOLO across all core metrics remain at a low level, indicating its strong robustness. This fully demonstrates that the proposed framework is an efficient solution.

a) Fairness and reliability analysis of comparative experiments: To ensure the fairness and objectivity of the performance evaluation, potential sources of unfairness for the directly cited data in Table V are discussed, including input resolution, training epochs, data augmentation strategies, pretraining, and slicing strategies. The summarized results are presented in Table VI.

First, regarding the difference in input resolution, larger input resolutions typically yield significant accuracy improvements. AEC-YOLO and the reproduced models strictly adopt a 640×640 resolution. In contrast, models such as QueryDet [54] use both high-resolution inputs and a slicing strategy, whereas CEASC [55] adopts a relatively high input resolution. These settings can improve detail preservation for small objects. Despite operating at a lower resolution, AEC-YOLO still outperforms these models. This verifies its capability to effectively extract features at low pixel densities.

Second, regarding the differences in training epochs and pretraining, the number of training epochs is typically carefully tuned by the original authors based on the convergence speed and overfitting risk of their model architectures. Meanwhile, pretrained weights endow models with robust general feature extraction capabilities, facilitating the rapid achievement of high accuracy. Our work adheres to the 300 epochs industry standard of the YOLO series without utilizing pretraining. TA-YOLO-s [61] was trained from scratch for 500 epochs to maximize model convergence. Conversely, although QueryDet [54] and CEASC [55] were trained for only 124 and 15 epochs, respectively, they rely on pretrained models. For certain lightweight models (such as EWS-YOLO [57]), 200 epochs might have already reached the saturation point of their feature expression capabilities. AEC-YOLO achieves stable and leading performance under the standard 300 epochs without using pretrained models, thereby striking an ideal balance between accuracy improvement and training costs.

Finally, regarding the differences in data augmentation and slicing strategies, our reproduced models uniformly adopt the Mosaic augmentation strategy, which is the standard configuration of the YOLO series. QueryDet [54] and CEASC [55] utilized random multiscale scaling and flipping. In comparison, the Mosaic augmentation strategy creates a more complex contextual environment. Furthermore, QueryDet [54] employed a slicing strategy, which partitions large images into overlapping patches for separate detection. This strategy significantly improves the accuracy of extremely small objects. AEC-YOLO achieves higher accuracy and inference speed without adopting a slicing strategy, demonstrating the effectiveness of the proposed improvements.

b) Efficiency–accuracy tradeoff and deployment analysis: To systematically evaluate the practicality of AEC-YOLO on low-altitude UAV platforms, Fig. 14 illustrates a scatter plot comparing the mAP_{50} and computational complexity (GFLOPs) of various models, comprehensively presenting the performance across three dimensions: accuracy, computational cost, and model scale.

As illustrated in Fig. 14, AEC-YOLO achieves the optimal tradeoff between computational cost and detection accuracy. Compared with the baseline model, although the computational complexity (GFLOPs) increases moderately, it yields

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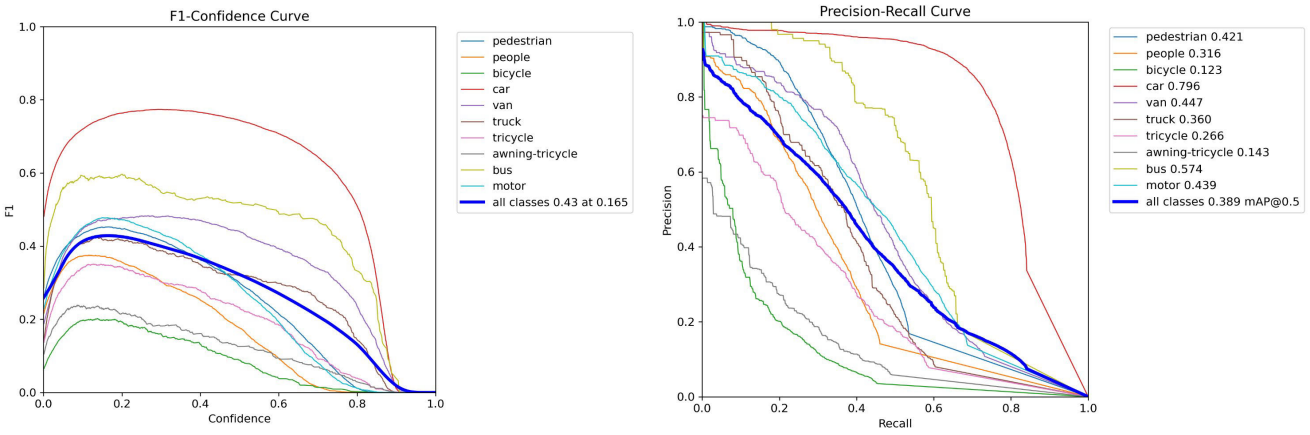


Fig. 12. Comprehensive performance curves of YOLO11s on the VisDrone2019 dataset.

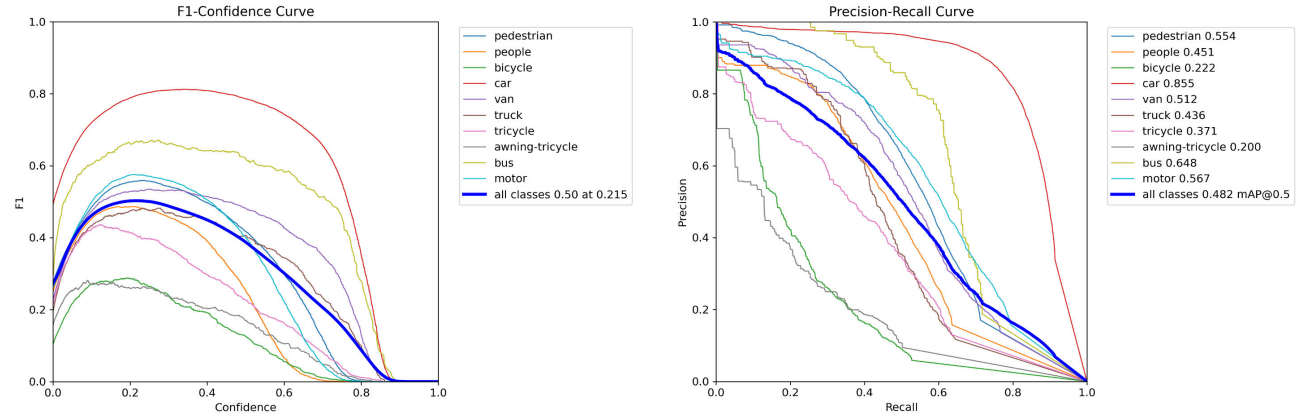


Fig. 13. Comprehensive performance curves of AEC-YOLO on the VisDrone2019 dataset.

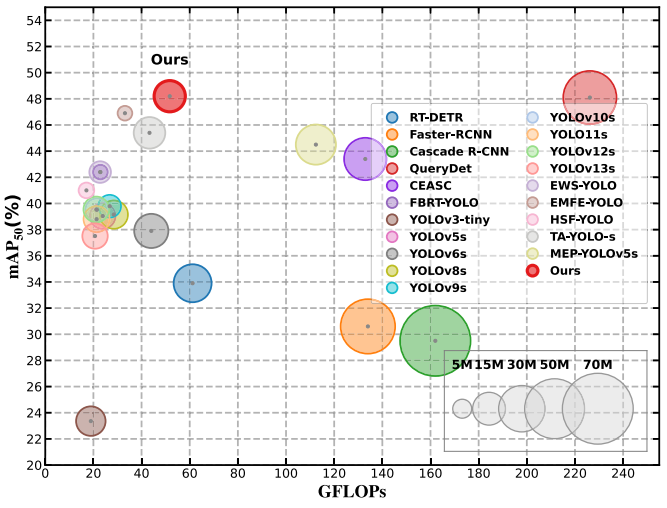
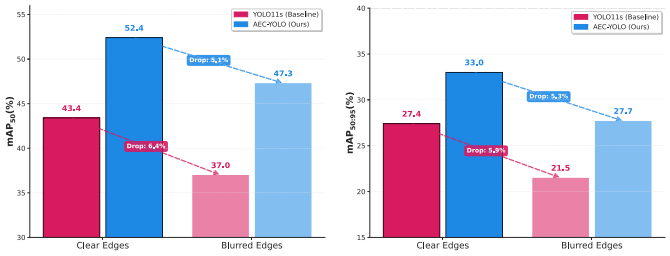


Fig. 14. Tradeoff between mAP_{50} and computational complexity on the VisDrone2019 dataset. The bubble size represents the number of parameters.

a significant accuracy improvement (+9.43%). Compared with SOTA methods, AEC-YOLO achieves leading detection performance while maintaining a moderate computational overhead suitable for low-altitude UAV platforms.

In terms of computational efficiency and real-time performance, AEC-YOLO demonstrates favorable engineering deployment feasibility. The model inference FPS reaches 61, satisfying the 30 FPS real-time requirement [52], [53]. In contrast, although models like EWS-YOLO [57] achieve higher frame rates, they often sacrifice the recall rate of small objects. Furthermore, constrained by the sampling frequency of airborne hardware, they are prone to cause computational power waste. For edge deployment on resource-constrained platforms (such as the NVIDIA Jetson series), where TensorRT is typically employed for FP16 or INT8 quantization [65], AEC-YOLO exhibits hardware-friendly inference characteristics. Quantized inference indicates that when processing a 51.7 GFLOPs computational task on an NVIDIA Jetson Orin NX (70–100 TOPS), the theoretical single-frame inference latency can be stably controlled within 20 ms after TensorRT FP16 quantization acceleration. This satisfies the real-time requirements of exceeding 30 FPS and maintaining a latency of under 50 ms. This performance stems from the IBN [49] structure adopted by the core component BAC-Net. This structure exhibits excellent computational utilization under acceleration frameworks like TensorRT, making the proposed framework highly compatible with hardware acceleration tools such as TensorRT on resource-constrained platforms. In

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(a) (b)

Fig. 15. Performance comparison between the baseline model and AEC-YOLO on subsets with different edge qualities. (a) mAP_{50} comparison. (b) $mAP_{50:95}$ comparison. The blue and red dashed lines represent the performance degradation of the two models as image quality decreases, quantitatively demonstrating the superiority of the proposed method in handling targets with blurred edges.

summary, AEC-YOLO successfully balances high-precision detection and efficient edge inference, possessing reliable real-time inference capabilities and deployment feasibility.

3) Performance Comparative Analysis Across Different Edge Quality Subsets: To investigate whether the optimization of edge information acts as the core driving factor for the performance improvement of AEC-YOLO, this section conducts a quantitative evaluation on different edge quality subsets. The specific experimental method is as follows. Based on an objective image evaluation standard, Laplacian variance is employed to calculate the richness of high-frequency edge information within the target regions of the validation set images [66]. According to the Laplacian scores, the samples are sorted in descending order. The top 30% and bottom 30% of the samples are then selected and partitioned into two subsets: the "clear edge" subset and the "blurred edge" subset. Subsequently, the detection performances of the baseline YOLO11s and AEC-YOLO are evaluated on these two subsets, utilizing mAP_{50} (%) and $mAP_{50:95}$ (%) as the core metrics. The experimental results are illustrated in Fig. 15.

As illustrated in Fig. 15, the experimental results demonstrate the edge information optimization capabilities of AEC-YOLO from two perspectives: performance gain and performance degradation control. First, regarding performance gain, AEC-YOLO achieves more significant improvements on edge-blurred samples. Under the mAP_{50} metric, AEC-YOLO improves by 9.0 percentage points on the edge-clear subset and an impressive 10.3 percentage points on the edge-blurred subset compared to the baseline model, with $mAP_{50:95}$ following the same trend. These results indicate that AEC-YOLO possesses stronger compensation capability when edge information is degraded. Second, in terms of performance degradation control, AEC-YOLO exhibits robust anti-interference capability. From edge-clear to edge-blurred scenarios, the performance drop of the baseline model under the mAP_{50} metric reaches 6.4% (red dashed line, left), whereas the drop for AEC-YOLO is suppressed to 5.1% (blue dashed line, left). For the $mAP_{50:95}$ metric, the decline is similarly reduced from 5.9% to 5.3% (right). This demonstrates that AEC-YOLO effectively mitigates the information loss caused by boundary blurring through its active calibration mechanism. In summary, quantitative experiments on subsets with different edge qualities confirm that the active perception and protection of edge information are the core drivers behind the performance breakthrough of AEC-YOLO.

D. Visual Analysis

1) Visualization of Detection Results: To intuitively evaluate the detection capability of AEC-YOLO in extremely complex scenarios, we selected three representative UAV-perspective challenges from the dataset for qualitative comparative analysis (see Fig. 16): severely blurred object boundaries, severe occlusion and densely overlapping objects, and extremely small and low-contrast objects. The yellow solid ellipses in the figure denote regions where AEC-YOLO significantly reduces missed detections compared with the baseline model. Meanwhile, the red dashed ellipses indicate failure cases of missed detections that still occur under physical limitations.

The results indicate that AEC-YOLO exhibits superior robustness over the baseline model in most challenging scenarios. When facing motion blur or dense object overlapping, the baseline model easily loses object contours, leading to massive missed detections. In contrast, benefiting from the capture and enhancement of high-frequency edge details by PERA and the precise spatial alignment of multiscale features by AECM, AEC-YOLO can effectively locate hard examples amidst complex backgrounds. Furthermore, the model achieves effective recall in low-contrast nighttime scenarios with extremely small objects, verifying the effectiveness of the proposed collaborative edge calibration strategy.

However, the model still exhibits limitations under extreme physical conditions (as indicated by the red dashed ellipses in Fig. 16). In scenarios with severely blurred boundaries, the extensive pixel smearing caused by high-speed UAV movement severely degrades the original high-frequency gradient information at the imaging level. This degradation exceeds the compensation capacity of our feature calibration mechanism. In severely occluded and densely overlapping regions, the boundary features of adjacent objects suffer from physical aliasing. Consequently, highly overlapping prediction boxes struggle to be independently retained during the NMS stage, causing missed detections of partially occluded instances. In addition, for extremely small and low-contrast objects, the virtual absence of extractable shallow edge signals prevents deep semantics from receiving effective detailed guidance, which similarly leads to perception failures.

In summary, despite certain limitations under extreme degradation conditions that exceed physical perception limits, AEC-YOLO achieves substantial improvements in detection accuracy and detail perception over the baseline model across the vast majority of real-world challenging scenarios.

2) Edge Feature Visualization: This section analyzes the performance of AEC-YOLO in extreme scenarios, such as occlusion, motion blur, and low lighting, by comparing the feature maps from the same layers of the YOLO11s and AEC-YOLO backbones. The results are illustrated in Fig. 17.

The analysis indicates that YOLO11s suffers from high-frequency edge degradation, where boundaries become blurred and entangled with the background. In contrast, AEC-YOLO

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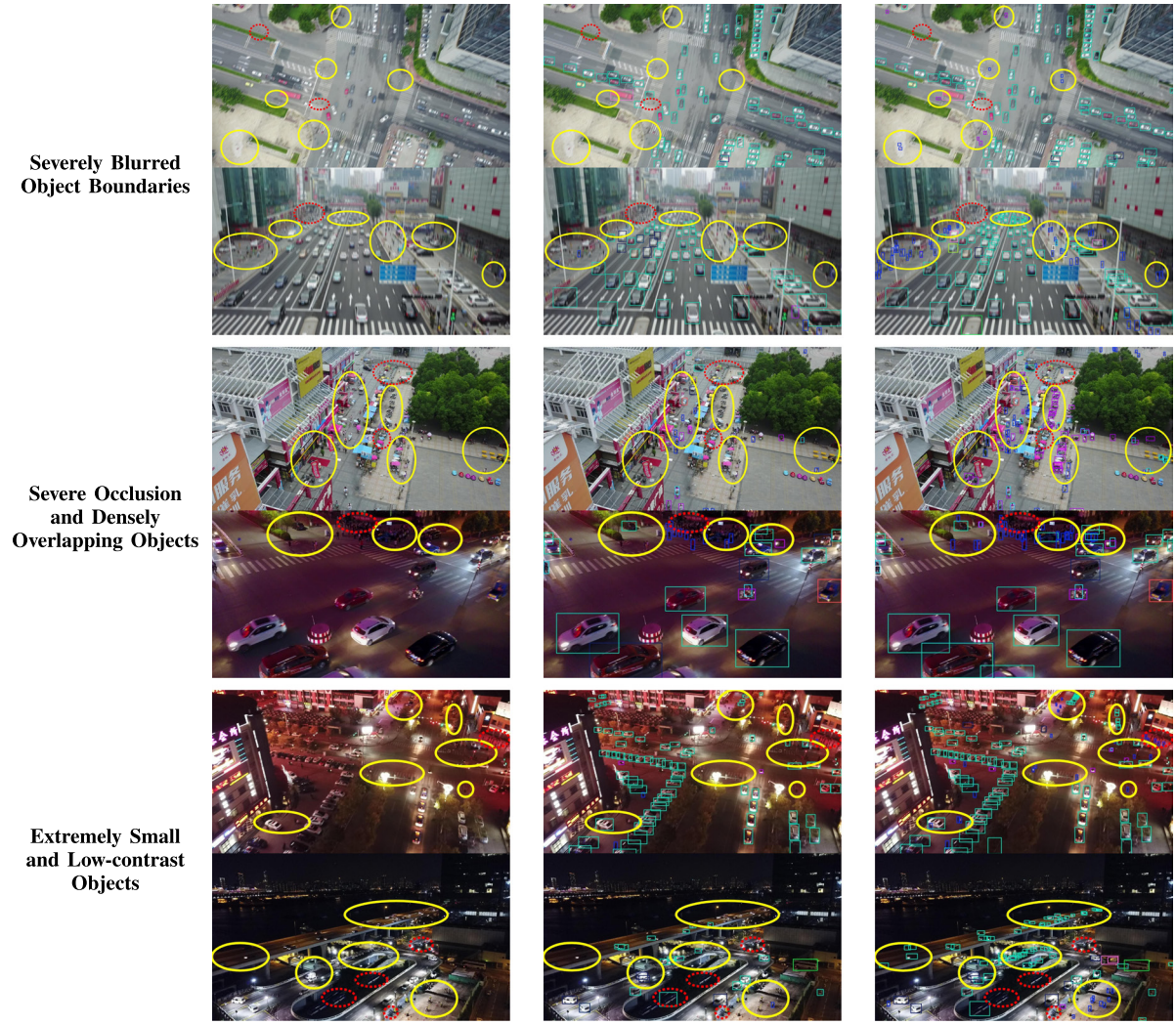


Fig. 16. Qualitative evaluation of detection results on the VisDrone2019 dataset under three extreme challenges. Yellow solid ellipses highlight regions where AEC-YOLO successfully reduces missed detections, while red dashed ellipses indicate failure cases where missed detections still occur. (a) Original images. (b) YOLO11s (Baseline). (c) AEC-YOLO (Ours).

maintains robustness across various scenarios: it effectively locks onto residual object contours in occluded scenes and suppresses noise interference under motion blur and low-light conditions, resulting in concentrated and clear responses for object contours (e.g., vehicle edges). These results confirm that AEC-YOLO effectively preserves high-frequency details and filters background interference through active feature refinement, providing high-quality feature representations for subsequent feature calibration and high-precision detection.

3) Comparative Analysis of Feature Evolution: This section compares and analyzes the feature evolution paths of AEC-YOLO and YOLO11s. The results are illustrated in Fig. 18. In the backbone network, high-frequency edge information of targets in the baseline model degrades, and edges become blurred. Conversely, PERA in AEC-YOLO reinforces edge responses in the early layers (Backbone-Layer 2). In deeper layers (Layers 4 and 6), its features remain clear and coherent, while those of the baseline model are scattered. At the functionally equivalent P3-scale feature fusion stage, the baseline model exhibits spatial aliasing and significant background clutter. Through BAC-Net, the AECM in AEC-YOLO achieves precise spatial alignment between high-level semantics and low-level details, ensuring that features are highly focused on the target core. This evolution process confirms that AEC-YOLO effectively preserves high-frequency edge information, providing a robust feature foundation for subsequent detection.

4) Feature Map Visualization for AECM: To intuitively verify the calibration effect of AECM, this section visualizes and compares the feature responses with and without AECM (taking the P3 layer as an example). As shown in Fig. 19, the color transitioning from blue to red in columns (b) and (c) represents increasing feature response intensity. The comparison shows that without AECM, due to feature misalignment, the object activations are highly dispersed and tend to diffuse into shadows and background regions. Consequently, the edge responses of objects are not concentrated, and the model is easily disturbed by background clutter. With AECM integrated, the high-response regions clearly focus on objects such as vehicles. The response boundaries between objects become sharper, and background interference is effectively

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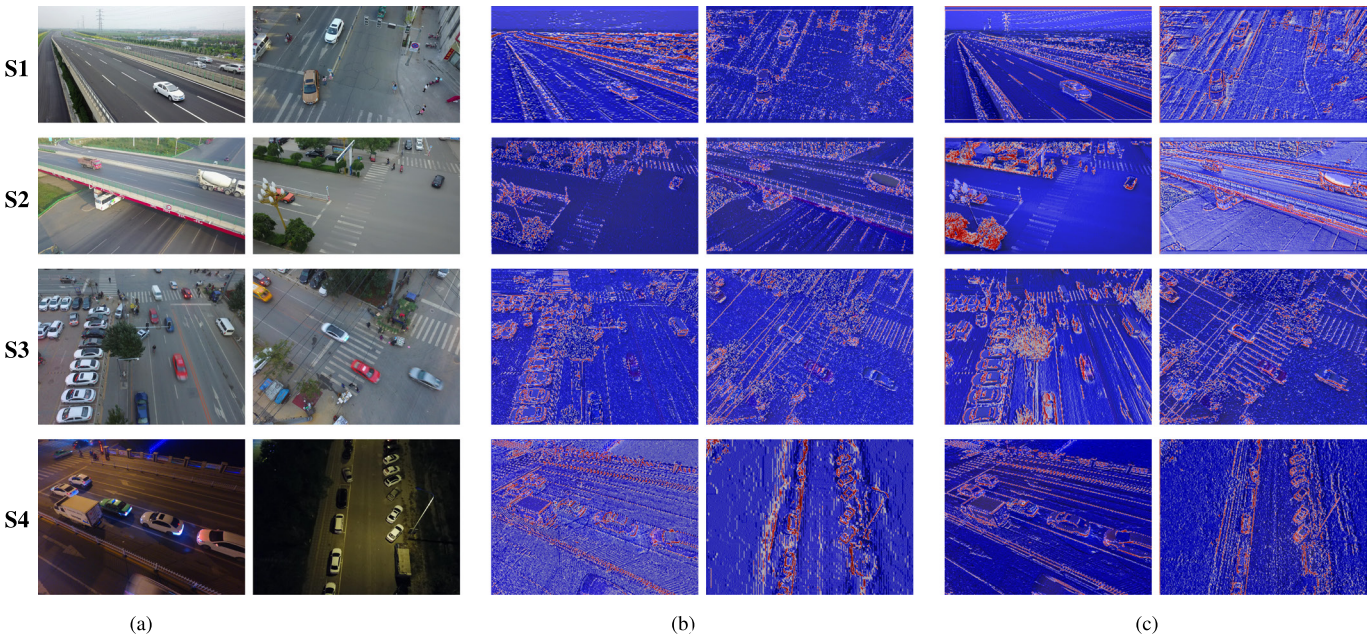


Fig. 17. Visual comparison of edge feature maps across four challenging scenarios: S1 (normal), S2 (occlusion), S3 (motion blur), and S4 (low light). (a) Original input images. (b) Feature maps generated by the baseline model YOLO11s (Baseline). (c) Feature maps generated by our proposed AEC-YOLO (Ours). All feature maps are extracted from equivalent backbone layers for a fair comparison.

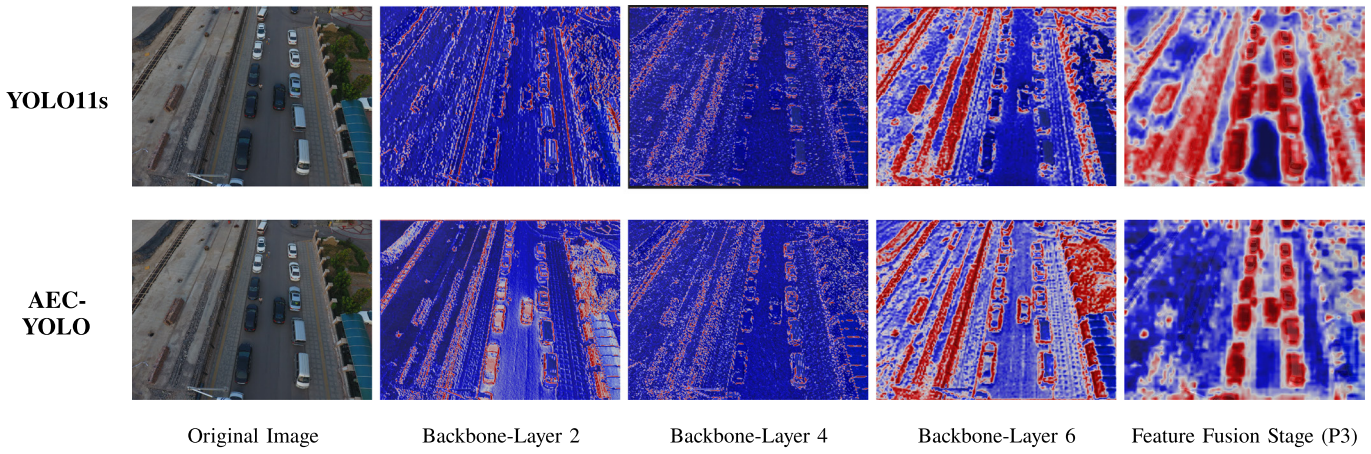


Fig. 18. Visual comparison of edge feature evolution between the baseline model and the proposed AEC-YOLO. The first column shows the original input image, followed by feature maps extracted from different stages of the network. The last column shows the output activation maps from the functionally equivalent feature-fusion stage of the two models at the P3 scale.

weakened. The difference map in column (d) further indicates that the red enhanced regions are mainly concentrated around object structures, whereas the blue suppressed regions are mostly distributed in nonobject background noise areas. These results demonstrate that AECM effectively calibrates the spatial misalignment between deep and shallow features, thereby improving the quality of small object feature representations.

5) Dynamic Loss Visualization: To evaluate the dynamic behavior and statistical stability of $\mathcal{L}_{\text{Adaptive}}$, the variation trends of the fusion ratio α were visualized through multiple independent observations, as shown in Fig. 20. To ensure the reliability of the experimental conclusions and mitigate the effects of random initialization, the figure illustrates the mean curve and standard deviation distribution of α from three independent training sessions: the dark cyan solid line represents the mean (Mean α), and the light blue shaded region denotes the standard deviation ($\pm\sigma$). The partial enlargement in the bottom-right corner further indicates that the model exhibits extremely low volatility and high consistency during the convergence stage, validating the robustness of the adaptive scheduling strategy under random initialization.

Two distinct stages are clearly observable. During the initial training stage (0–20 epochs), the model's localization accuracy is relatively low, with the batch average IoU ($\overline{\text{IoU}}$) remaining at a low level. Consequently, α rises rapidly from near 0.35. At this phase, the weight $(1-\alpha)$ dominates. The total loss function biases toward MPDIoU Loss [16], compelling the model to prioritize learning the geometric attributes of objects to achieve coarse localization. As training progresses, the model's localization capability is significantly enhanced,

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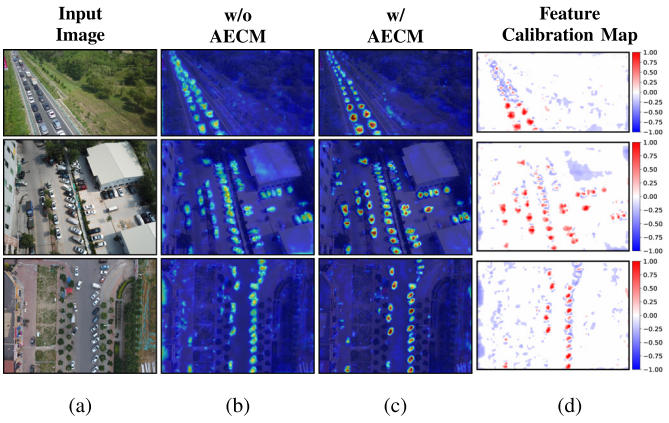


Fig. 19. (a) Input images. (b) Feature activation maps without AECM. (c) Feature activation maps with AECM. (d) Feature calibration difference maps (obtained by subtracting (b) from (c), where red regions represent the focused features enhanced by AECM, and blue regions represent the suppressed background and shadow noise). Warm colors indicate stronger activation responses, while cool colors indicate weaker activation responses.

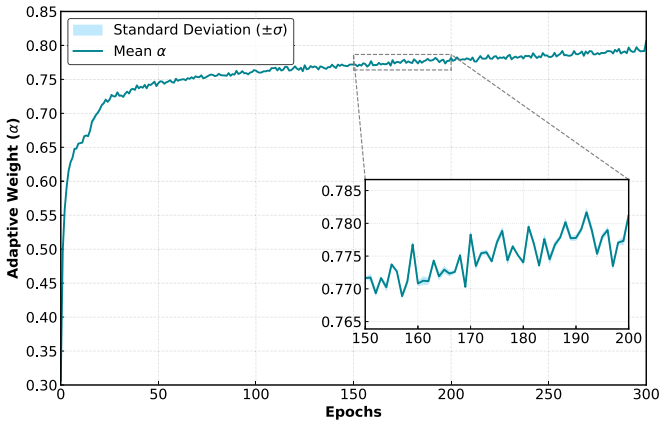


Fig. 20. Visualization of the statistical stability for the dynamic weight evolution of $\mathcal{L}_{\text{Adaptive}}$. The shaded area represents the standard deviation ($\pm\sigma$), and the solid line denotes the mean value of α.

and the $\overline{\text{IoU}}$ rises and maintains at a high level. Accordingly, α converges and stabilizes at approximately 0.8. During this phase, the weight α becomes dominant. The total loss function shifts toward Focaler-IoU Loss [15], prompting the model to optimize focused samples based on high IoU values and perform fine-grained tuning of object boundaries. These visualization results demonstrate that the adaptive strategy effectively guides the model through a smooth transition from coarse localization to fine-grained refinement.

E. Quantitative and Qualitative Evaluation of Illumination Sensitivity

To further quantitatively evaluate the sensitivity of the model to illumination changes and verify its specific capability to utilize sparse lighting cues at night, an extended evaluation was conducted. To ensure objectivity, an automated script was developed to partition the VisDrone2019 validation set.

Specifically, the average value $\overline{V}$ of the value (V) channel in the HSV color space was calculated for each image. The threshold was set to $T_v = 75$ (range of 0–255), where images

TABLE VII

QUANTITATIVE PERFORMANCE COMPARISON UNDER DIFFERENT ILLUMINATION SCENARIOS

Model	Scenario	Precision (%)	Recall (%)	mAP_{50} (%)	$\text{mAP}_{50:95}$ (%)
YOLO11s (Baseline)	Daylight	49.66	38.58	38.79	23.47
	Night	55.04	39.25	43.55	26.69
AEC-YOLO (Ours)	Daylight	**58.03**	**45.78**	**48.22**	**29.77**
	Night	**62.77**	**46.67**	**51.76**	**31.99**

Note: **Bold** values indicate the best results for each illumination scenario.

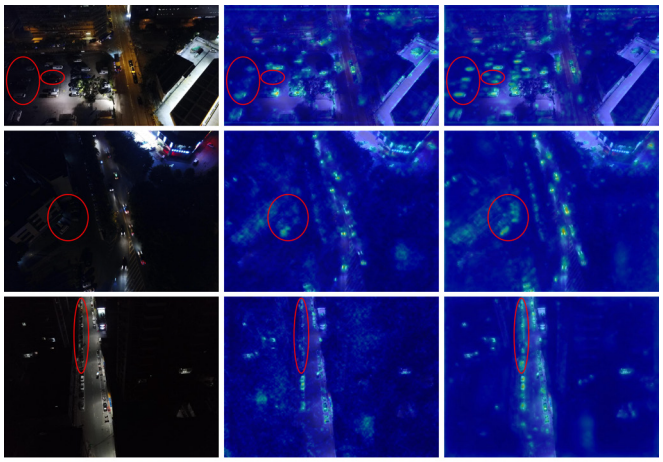


Fig. 21. Visual comparison of feature activation in nighttime scenes. (a) Original input images. (b) Feature maps generated by the baseline YOLO11s. (c) Feature maps generated by the proposed AEC-YOLO. The color scale from dark blue to bright yellow indicates the intensity of the feature response. Red circles highlight key regions where targets are severely occluded by extremely low light.

with $\overline{V} < T_v$ were defined as the night subset and the rest as the day subset. Subsequently, the baseline model YOLO11s and the proposed AEC-YOLO were evaluated directly on these two subsets.

As illustrated in Table VII, the evaluation results demonstrate that AEC-YOLO consistently surpasses the baseline model across diverse illumination conditions. Specifically, the precision in the night subset reaches 62.77%, achieving the highest global accuracy. In nighttime scenes, the low-frequency textures of objects typically undergo degradation, leaving only faint high-frequency boundary signals—such as subtle reflections or light-source contours—as effective physical cues. By actively capturing these high-frequency signals via the PERA architecture and calibrating sparse cues through the AECM, AEC-YOLO achieves a recall of 46.67% (+7.42%) and an mAP_{50} of 51.76% (+8.21%) in nighttime conditions. This effectively minimizes the missed detection rate and enables the proposed model to identify a greater number of hard examples. The simultaneous improvement in both precision and recall within the nighttime subset quantitatively validates the capacity of AEC-YOLO to exploit sparse illumination cues, thereby significantly enhancing the detection performance for faint objects.

Fig. 21 presents a visual comparison of feature activation in extreme low-light nighttime scenes. The color scale from dark blue to bright yellow indicates the intensity of the network feature response. In the low-light occluded areas marked by

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TABLE VIII

PERFORMANCE COMPARISON OF AEC-YOLO WITH OTHER OBJECT DETECTION MODELS ON THE AI-TOD DATASET

Model	AP	BR	ST	SH	SP	VE	PE	WM	P	R	mAP_{50}	$mAP_{50:95}$	Param	FLOPs	FPS	F1
Faster R-CNN†	28.3	17.5	44.9	38.1	5.3	25.4	11.8	0.1	35.2	25.4	21.4	9.6	41.4	134.0	42.0	29.5
Cascade R-CNN†	34.4	26.1	28.1	38.6	15.1	27.3	10.5	0.8	37.1	26.8	22.6	10.9	69.2	162.0	21.0	31.1
RetinaNet†	18.4	25.7	30.7	45.7	4.0	25.2	11.5	0.0	33.8	28.1	20.2	9.4	36.5	133.0	45.0	30.7
YOLOv3-tiny†	34.2	11.0	50.7	43.5	5.0	37.8	10.9	5.6	36.4	29.5	24.8	10.2	12.1	18.9	138.0	32.6
YOLOv5s†	52.6	37.9	77.2	65.6	9.0	69.9	24.6	8.6	51.6	44.7	43.2	19.2	9.1	23.9	113.0	47.9
YOLOv6s†	41.7	26.4	73.6	62.1	2.2	66.8	20.8	1.0	45.3	39.2	36.8	16.3	16.3	44.0	63.0	42.0
YOLOv8s†	**58.7**	36.7	80.5	66.1	18.4	70.5	25.1	10.2	54.1	46.5	45.8	20.4	11.1	28.5	99.0	50.0
YOLOv9s†	53.5	31.8	77.1	65.9	9.6	69.6	24.3	11.6	52.8	43.4	42.9	19.0	7.2	26.8	107.0	47.6
YOLOv10s†	56.7	34.8	76.7	65.2	14.3	69.6	25.2	12.9	53.4	45.1	44.4	20.0	7.2	21.4	134.0	48.9
YOLO11s†	50.4	35.4	76.0	63.3	**21.0**	67.4	23.4	10.4	52.5	44.3	43.3	19.1	9.4	21.3	128.0	48.1
YOLOv12s†	48.2	28.6	76.7	64.4	19.3	69.7	23.4	5.2	51.1	42.8	41.9	18.7	9.2	21.2	134.0	46.6
YOLOv13s†	43.3	36.5	80.0	65.7	19.2	68.8	23.2	12.3	52.7	44.6	43.6	19.2	9.0	20.7	136.0	48.3
AEC-YOLO†	37.8	**51.1**	**84.8**	**70.4**	19.5	**73.3**	**28.8**	**19.5**	**57.3**	**48.6**	**48.2**	**21.7**	13.6	51.7	57.0	**52.6**

Note: † indicates models implemented in our local environment following a uniform experimental setup. **P**, **R**, and **F1** denote Precision, Recall, and F1-score, respectively. **Params** (in Millions, M) and **FLOPs** (in Giga, 10^9) represent the number of parameters and floating-point operations. FPS stands for frames per second. All performance metrics (category-specific AP, P, R, mAP, and F1) are reported as percentages (%). These notations and units apply consistently to Tables IX and X. Category abbreviations: AP: Airplane, BR: Bridge, ST: Storage Tank, SH: Ship, SP: Swimming Pool, VE: Vehicle, PE: Person, WM: Windmill. Best results are highlighted in **bold**.

TABLE IX

PERFORMANCE COMPARISON BETWEEN THE PROPOSED AEC-YOLO AND OTHER OBJECT DETECTION MODELS ON THE DOTA-V1.5 DATASET

Model	PL	SH	ST	BD	TC	BC	GTF	HA	BR	LV	SV	HC	RA	SBF	SP	CC	P	R	mAP_{50}	$mAP_{50:95}$	Param	FLOPs	FPS	F1
Faster R-CNN†	60.3	47.1	31.0	42.5	73.7	28.7	30.6	56.1	11.5	52.9	20.3	24.0	13.5	24.0	51.3	0.0	62.5	37.8	35.5	21.3	41.4	134.0	21.0	47.1
Cascade R-CNN†	55.8	43.5	28.6	39.2	68.1	26.5	28.2	51.9	10.6	48.9	18.7	22.1	12.5	22.2	47.4	0.0	58.7	35.1	32.8	20.5	69.2	162.0	16.0	43.9
RetinaNet†	33.0	25.7	16.9	23.2	40.3	15.7	16.7	30.6	6.3	28.9	11.1	13.1	7.4	13.1	28.0	0.0	46.8	22.3	19.4	11.4	36.5	133.0	24.0	30.2
YOLOv3-tiny†	51.4	25.3	9.2	34.9	89.0	28.7	30.1	49.8	2.5	53.2	10.1	17.5	7.6	30.4	14.6	0.0	56.4	31.2	28.4	17.1	12.1	18.9	152.0	40.2
YOLOv5s†	70.1	51.6	28.9	41.0	92.0	38.3	30.4	64.5	6.4	69.8	20.8	27.9	12.3	36.0	38.3	0.0	67.2	41.5	39.3	24.3	9.1	23.9	118.0	51.3
YOLOv6s†	69.0	48.8	26.4	37.3	91.9	35.3	31.7	63.8	6.8	66.6	18.8	22.8	9.9	32.4	34.1	0.0	71.6	39.4	37.2	17.1	16.3	44.0	72.0	50.8
YOLOv8s†	69.8	51.1	29.4	41.5	91.6	**39.7**	34.8	67.6	8.3	69.8	20.7	25.8	14.0	**39.5**	38.1	0.1	66.8	42.3	40.1	24.9	11.1	28.5	105.0	51.8
YOLOv9s†	70.4	50.4	26.4	44.0	**92.5**	37.6	37.1	67.7	7.8	69.9	20.3	24.3	13.6	37.3	40.0	0.0	68.1	42.0	40.0	23.0	7.2	26.8	112.0	52.0
YOLOv10s†	67.8	50.7	27.0	42.8	91.4	36.4	30.2	60.2	8.2	67.5	20.2	28.8	11.3	35.0	37.2	0.0	62.9	40.6	38.4	24.3	7.2	21.4	142.0	49.3
YOLO11s†	70.1	51.2	28.5	40.8	91.8	38.0	30.2	64.2	6.5	69.5	20.6	27.5	12.0	35.8	42.1	0.0	71.3	41.2	39.3	24.6	9.4	21.3	138.0	52.2
YOLOv12s†	69.6	51.6	28.2	40.7	92.0	38.0	37.0	64.5	8.3	69.6	20.1	27.2	10.8	36.8	38.2	0.0	63.0	41.6	39.5	24.9	9.2	21.2	140.0	50.1
YOLOv13s†	67.7	50.5	29.0	39.3	91.0	35.1	32.1	64.6	5.4	69.3	19.8	26.3	12.9	31.8	36.3	0.1	64.6	40.3	38.2	25.1	9.0	20.7	132.0	49.6
AEC-YOLO†	**74.4**	**58.6**	**38.7**	**52.9**	92.1	35.9	**38.3**	**70.1**	**14.5**	**71.2**	**25.4**	**30.0**	**17.0**	30.1	**53.5**	**0.1**	**75.8**	**45.2**	**43.9**	**26.8**	13.6	51.7	59.0	**56.6**

Note: For the detailed definitions of evaluation metrics, unit specifications, and other related explanations, please refer to the Note in Table VIII. PL: Plane, SH: Ship, ST: Storage Tank, BD: Baseball Diamond, TC: Tennis Court, BC: Basketball Court, GTF: Ground Track Field, HA: Harbor, BR: Bridge, LV: Large Vehicle, SV: Small Vehicle, HC: Helicopter, RA: Roundabout, SBF: Soccer Ball Field, SP: Swimming Pool, CC: Container Crane.

TABLE X

PERFORMANCE COMPARISON BETWEEN THE PROPOSED AEC-YOLO AND OTHER OBJECT DETECTION MODELS ON THE DIOR DATASET

Model	AP	AIP	BF	BC	BR	CH	DA	ESA	ETS	GF	GTF	HA	OP	SH	STA	ST	TC	TS	VE	WM	P	R	mAP_{50}	$mAP_{50:95}$	Param	FLOPs	FPS	F1
Faster R-CNN†	91.1	67.8	91.9	82.3	38.7	90.8	53.2	60.9	57.0	75.8	45.2	71.0	60.7	77.5	86.7	80.4	87.8	47.9	70.6	73.7	78.2	73.5	70.6	46.2	41.4	134.0	35	75.8
Cascade R-CNN†	90.9	67.6	91.9	80.8	40.6	90.8	52.2	64.6	56.3	77.2	52.3	75.5	61.6	77.6	89.4	78.8	87.0	45.8	69.0	75.8	79.4	74.1	71.3	47.5	69.2	162.0	18	76.7
RetinaNet†	91.6	71.0	92.1	81.6	36.4	89.9	47.6	65.1	53.7	74.8	52.1	74.2	59.4	73.7	80.3	81.3	88.3	49.1	67.1	74.6	77.6	72.8	70.2	45.8	36.5	133.0	32	75.1
YOLOv3-tiny†	92.2	65.8	93.5	82.4	41.1	88.9	53.1	69.7	63.6	66.9	71.2	65.5	63.4	80.8	94.0	79.8	92.3	48.0	62.8	77.5	80.3	74.9	72.6	48.1	12.1	18.9	145	77.5
YOLOv5s†	92.2	78.5	90.4	80.1	47.5	88.2	65.5	72.1	65.4	78.8	80.2	72.4	63.8	91.2	90.8	81.1	91.0	60.1	75.2	84.5	83.5	79.2	77.4	53.6	9.1	23.9	115	81.3
YOLOv6s†	92.4	84.8	94.8	84.6	50.9	89.8	66.0	73.5	65.8	79.4	80.8	73.6	67.8	94.4	92.7	82.2	93.6	55.1	75.1	83.4	84.8	80.6	79.0	55.4	16.3	44.0	68	82.6
YOLOv8s†	94.5	84.2	94.9	84.2	50.1	91.3	69.8	76.9	66.8	81.3	**87.5**	73.2	68.0	94.5	**97.2**	83.0	92.7	67.3	75.7	82.9	86.3	82.1	80.8	58.2	11.1	28.5	102	84.1
YOLOv9s†	95.1	86.8	93.4	**86.4**	52.1	90.2	71.0	77.0	69.5	78.2	81.5	73.8	68.5	94.2	93.8	84.2	92.5	61.5	77.2	83.1	86.0	81.7	80.5	57.9	7.2	26.8	110	83.8
YOLOv10s†	95.3	85.3	95.2	85.9	52.1	90.3	66.7	77.3	68.9	76.4	80.3	72.6	67.8	95.1	93.4	84.5	93.1	63.1	77.9	83.5	85.6	81.3	80.2	57.5	7.2	21.4	138	83.4
YOLO11s†	95.2	85.0	**96.8**	83.5	49.5	89.5	61.5	74.2	69.8	76.5	80.2	71.5	66.2	93.8	93.5	83.2	92.0	60.5	75.2	82.4	84.5	80.3	79.0	57.2	9.4	21.3	132	82.3
YOLOv12s†	95.5	**87.8**	94.5	85.0	53.2	90.5	72.0	74.1	69.4	83.2	84.0	74.2	68.0	95.1	94.2	84.0	93.2	63.8	77.5	84.0	86.7	82.5	81.2	58.7	9.2	21.2	136	84.5
YOLOv13s†	94.3	84.2	94.5	84.9	52.2	91.3	67.1	75.1	70.8	79.1	81.8	74.1	65.7	94.7	93.9	84.4	93.3	56.8	77.8	83.5	85.4	81.1	80.0	56.8	9.0	20.7	134	83.2
AEC-YOLO†	**96.5**	86.2	95.5	85.1	**55.0**	**92.5**	**75.2**	**81.0**	**73.8**	**87.2**	86.5	**75.8**	**71.5**	**97.2**	95.8	**87.3**	**95.9**	**69.2**	**84.5**	**87.5**	**89.5**	**84.8**	**83.96**	**61.4**	13.6	51.7	55	**87.1**

Note: For the detailed definitions of evaluation metrics, unit specifications, and other related explanations, please refer to the Note in Table VIII. AP: Airplane, AIP: Airport, BF: Baseball Field, BC: Basketball Court, BR: Bridge, CH: Chimney, DA: Dam, ESA: Expressway Service Area, ETS: Expressway Toll Station, GF: Golf Field, GTF: Ground Track Field, HA: Harbor, OP: Overpass, SH: Ship, STA: Stadium, ST: Storage Tank, TC: Tennis Court, TS: Train Station, VE: Vehicle, WM: Windmill.

red circles, the baseline YOLO11s model shows very weak and scattered activation, making it difficult to effectively focus on targets amid background noise. In contrast, benefiting from the active edge preservation and collaborative feature calibration of the PERA and AECM modules, our network displays highly concentrated activation in the same areas, appearing as warm-colored patches with high response. This qualitative comparison shows that AEC-YOLO can capture

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Fig. 22. Visual comparison of detection results on AI-TOD, DOTA-v1.5, and DIOR. Yellow ellipses mark hard regions where AEC-YOLO reduces missed detections and improves localization under complex backgrounds and tiny-object conditions.

sparse illumination cues in nighttime environments, strongly demonstrating its robustness to lighting variations.

F. Generalization Analysis on Multiple Benchmarks

This section validates the generalization of AEC-YOLO through quantitative and qualitative experiments on three high-difficulty remote sensing datasets: AI-TOD [18], DOTA-v1.5 [19], and DIOR [20]. These datasets encompass challenges such as tiny objects, complex noisy backgrounds, and category diversity. To rigorously verify the generalization capability of AEC-YOLO, the optimal hyperparameter combination ($\beta = 0.9, u = 0.8$) obtained from VisDrone2019 was directly adopted for $\mathcal{L}_{\text{Adaptive}}$ in the following experiments.

The quantitative analysis results on the AI-TOD [18], DOTA-v1.5 [19], and DIOR [20] datasets are presented in Tables VIII–X, respectively. The proposed AEC-YOLO achieves consistent improvements across multiple benchmarks. On AI-TOD [18] (targeting extremely tiny objects), AEC-YOLO yields 48.2% mAP_{50} (+4.9 percentage points over baseline). Its precision (57.3%), recall (48.6%), and $F1$-score

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(52.6%) surpass YOLO11s by 4.8, 4.3, and 4.5 percentage points, respectively. This confirms the framework effectively balances FP suppression and missed detection rates. Category-level analysis reveals that the model achieves significant gains in categories characterized by distinct rigid contours. For instance, the AP for bridge (BR, 51.1%), storage tank (ST, 84.8%), and wind mill (WM, 19.5%) improves by 15.7, 8.8, and 9.1 percentage points over the baseline, respectively. In contrast, the improvements are limited for categories like swimming pool (SP), which lack regular edges and are prone to boundary blending with the background. This performance discrepancy further validates the capability of the PERA module to capture the sparse geometric edges of tiny objects. It firmly demonstrates that edge extraction is the core mechanism driving the model's performance improvements.

Second, on DOTA-v1.5 [19] (featuring rich categories and drastic scale variations), AEC-YOLO leads with 43.9% mAP_{50} (+4.6 percentage points over baseline) and 26.8% $mAP_{50:95}$. Category-level analysis reveals that the model achieves prominent gains in categories with distinct edge structures, such as helicopter (HC, +9.4%) and baseball diamond (BD, +12.1%). This improvement is attributed to the AECM, which effectively calibrates the geometric priors of complex contours. Furthermore, the model demonstrates strong robustness on traditional hard examples [HC, small vehicle (SV), and soccer ball field (SBF)], achieving APs of 30.0%, 25.4%, and 30.1%, respectively. This distribution pattern of performance gains further validates the underlying design logic of our framework. The $F1$-score hits 56.6% (+4.4 percentage points over YOLO11s). This balanced growth in precision (75.8%) and recall (45.2%) indicates AEC-YOLO's superior discriminative capability to suppress FPs in noisy backgrounds.

Finally, on DIOR [20] (containing the most categories), AEC-YOLO outperforms YOLO11s in precision (89.5%), recall (84.8%), and $F1$-score (87.1%) by 5.0, 4.5, and 4.8 percentage points, respectively. Its mAP_{50} reaches an optimal 83.96% (+4.96 percentage points over baseline). Category-level analysis reveals that the model achieves accuracy improvements of 5.5, 13.7, and 5.3 percentage points over the baseline in complex geometric categories with large-span linear or irregular hard boundaries, such as bridge (BR), dam (DA), and overpass (OP), respectively. This demonstrates that the edge calibration mechanism effectively enhances the structural representations of such targets. This consistent distribution pattern of performance gains across datasets verifies that the model's generalization and robustness in complex remote sensing scenarios are built upon the precise capture and adaptive calibration of the physical edge attributes of targets. Across the three datasets, these data strongly demonstrate that our model maintains stable generalization capability on new and more challenging datasets.

To qualitatively verify the superiority of AEC-YOLO, Fig. 22 displays the comparative detection results on three benchmark datasets (AI-TOD [18], DOTA-v1.5 [19], and DIOR [20]). An in-depth analysis of the hard example regions marked by yellow ellipses reveals that AEC-YOLO exhibits superior capture precision and excellent robustness when processing objects with tiny scales, dense distributions, and high background interference.

In the tiny object scenarios of the AI-TOD dataset [18] (rows 1 and 2), RT-DETR [64] and YOLO11s [31] still suffer from missed detections. Although YOLOv12s [32] and YOLOv13s [33] show certain improvements, they still exhibit incomplete localization and missed detections under complex backgrounds. In contrast, AEC-YOLO detects more small targets within the yellow elliptical regions and achieves more accurate localization.

In complex remote sensing scenarios from DOTA-v1.5 [19] and DIOR [20] (rows 3–6), the yellow elliptical regions further highlight the limitations of baseline models in handling shadow occlusion, drastic scale variations, and low-contrast objects. Specifically, in the shaded areas of highway interchanges and dense port vessel regions, AEC-YOLO significantly reduces the missed detection rate through its improvements. The visual comparison results strongly demonstrate the adaptability of the model to extremely complex scenarios, further validating the superior generalization performance of the proposed framework.

V. Conclusion

To address core challenges such as edge information degradation, insufficient feature alignment, and localization sensitivity in small object detection within UAV aerial scenarios, this study proposes the AEC-YOLO detection framework. Substantial performance gains are realized through innovations in three key aspects.

1) The PERA structure is designed. By utilizing parallel branches and a progressive pyramid fusion strategy, it actively enhances high-frequency edge details to resolve the issue of feature blurring caused by downsampling.
2) The AECM mechanism is constructed. Combining the BAC-Net and a P2 detection head, it achieves collaborative optimization of deep semantics and shallow edges, significantly alleviating the semantic gap.
3) The adaptive weight Focaler-MPDIoU loss function is formulated. Through adaptive scheduling based on IoU during training, it balances the requirements for optimizing sample imbalance and localization precision.

Experiments on the VisDrone2019 [17] dataset indicate that AEC-YOLO improves mAP_{50} by 9.43% and $mAP_{50:95}$ by 6.45 percentage points compared to the baseline YOLO11s. In particular, it demonstrates gains of 13.5 and 12.8 percentage points for "people" and "motor," respectively. To evaluate the model's robustness across diverse scenarios, extensive generalization analyses were conducted on the AI-TOD [18], DOTA-v1.5 [19], and DIOR [20] datasets. The results show that AEC-YOLO consistently outperforms the baseline model, achieving mAP_{50} improvements of 4.9, 4.6, and 4.96 percentage points, respectively. Moreover, extensive quantitative comparisons verify that AEC-YOLO achieves leading advantages over numerous existing mainstream object detectors in terms of both detection accuracy and recall capabilities. This validates the effectiveness of the active edge information enhancement and adaptive feature calibration strategies. Future

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work will explore robustness optimization in dynamic occlusion scenarios and further compress computational overhead to more efficiently adapt to edge deployment requirements.

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